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POLITECNICO
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Machine Learning for Mechcanical Systems
Python Project

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A.Y. 2025 – 2026

Abstract

This report presents a surrogate-assisted optimization and validation workflow for a compliant base–arm mechanical system commanded through a chirp reference. The controlled output is the absolute arm displacement, $y = x_a + x_b$, and the objective is to exploit the coupled dynamics of the system to reach a prescribed target displacement larger than the maximum command amplitude. The design problem is treated as a constrained black-box optimization task: the command must reach the target band while keeping the absolute arm velocity, the actuation force and the dominant phase indicator within the adopted limits.

A simulation dataset was first generated from the true equations of motion and used to train a feed-forward Neural Network surrogate. A Physics-Informed Neural Network formulation was also investigated through an eigenvalue-residual loss, but the standard Neural Network provided a more reliable accuracy–simplicity trade-off and was therefore retained for the final optimization loop. Bayesian Optimization was then used to tune the arm parameters and chirp command variables, while all selected candidates were validated again with the true equations of motion.

The final nominal command reached the target band with high precision, giving $\max(y) = 0.150\,604\,\mathrm{m}$ for a target value of $0.150\,000\,\mathrm{m}$, with velocity and force remaining below their monitored limits. Robustness was assessed with respect to broad base-parameter perturbations, local arm-parameter variations and initial-condition perturbations. The results show that the nominal solution is dynamically effective but not uniformly robust: base damping and mass mainly affect reaching capability, arm damping strongly controls local robustness, and severe favourable initial conditions can produce target-band overshoot. A base-specific re-optimization demonstrated that the same surrogate-assisted workflow can recover admissible performance when the nominal command becomes marginal under perturbed-base conditions.

Contents

1 Introduction

1.1 Problem Statement and Project Objective

The objective of this project is to design and validate a reference command for a compliant base–arm system. The controlled quantity of interest is the absolute displacement of the arm, $y = x_a + x_b$, where x_a is the arm displacement relative to the base and x_b is the absolute displacement of the base. The target is not simply to track the imposed reference motion, but to exploit the coupled dynamics of the compliant system so that y reaches a prescribed target value larger than the maximum command amplitude.

This makes the problem intrinsically dynamic. A purely static or quasi-static command would not take advantage of the amplification introduced by the compliant base and by the arm dynamics. For this reason, the imposed reference is chosen as a chirp command, i.e., a sinusoidal signal whose frequency varies continuously over time. By tuning its lower frequency, upper frequency, duration and amplitude, the command can excite a frequency interval and interact with the natural modes of the coupled system.

However, reaching the target is not sufficient. The final command must also remain compatible with constraints on velocity, actuation force and target-band accuracy. The design problem is therefore treated as a constrained black-box optimization problem: each candidate command must be evaluated through a time-domain simulation, and several performance indicators must be extracted from the resulting response.

A Neural Network (NN) surrogate is introduced to make the optimization computationally feasible. The NN approximates the relevant simulation outputs from the mechanical parameters and the chirp parameters, and is then used as a fast evaluator inside the Bayesian Optimization (BO) loop. The surrogate does not replace the physical model in the final assessment: the selected candidates are re-integrated with the true Equations of Motion (EoM), and all final conclusions are based on true-EoM validation.

1.2 Dynamic Model Description

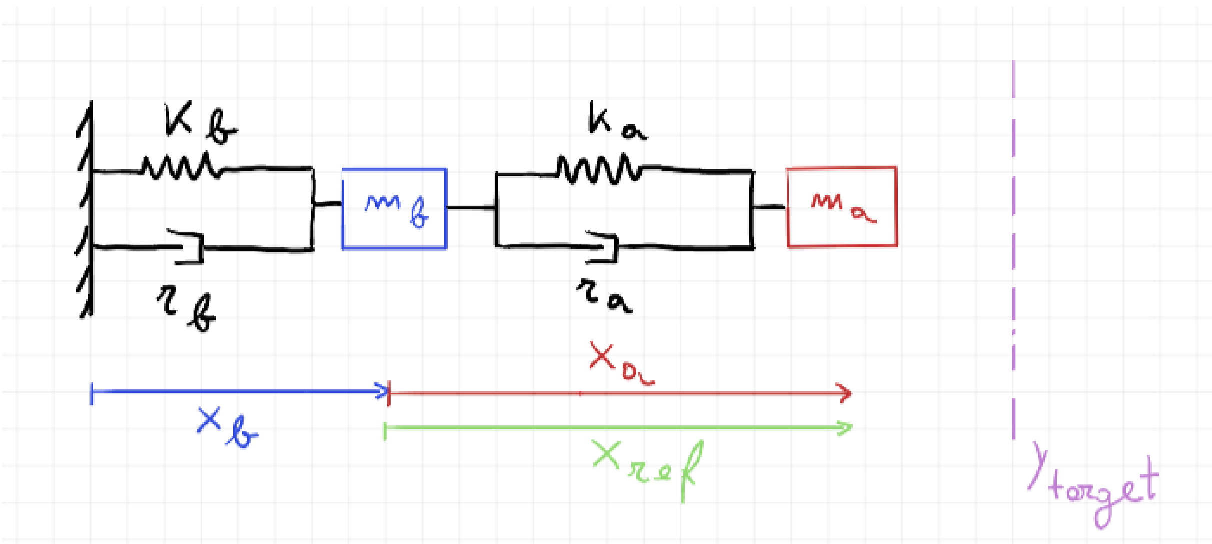

Figure 1: System model.

The simplified two-degree-of-freedom model is consistent with the compliant-base interaction framework discussed in the reference paper, which was used as a methodological guideline. The present work, however, does not aim to reproduce the controller proposed in that paper. Instead, the model is used as a simulation environment for dataset generation, surrogate modelling, Bayesian optimization and robustness validation.

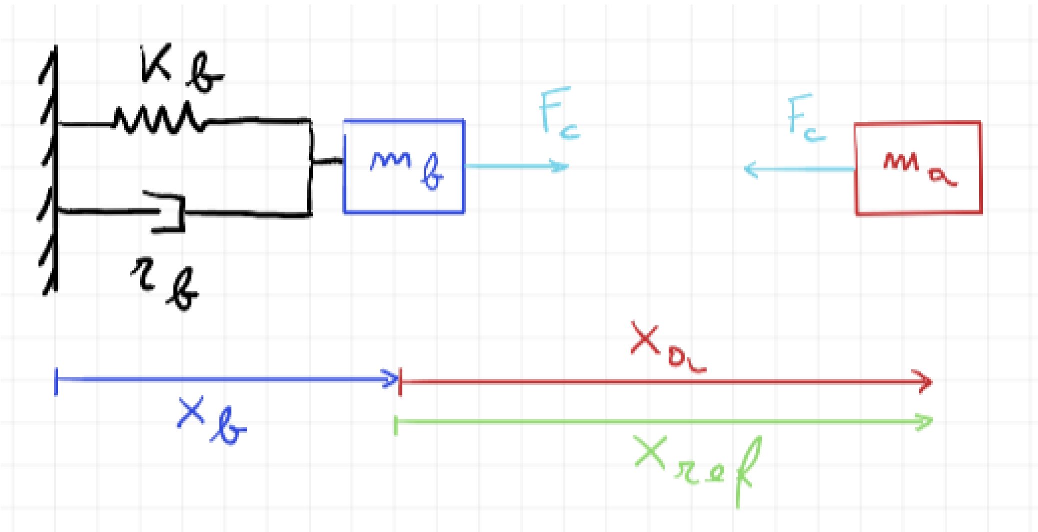

Figure 2: Control force acting on the system.

A consistent coordinate convention was fixed before proceeding with the surrogate model. The base displacement is denoted as x_b and is an absolute displacement measured with respect to a fixed inertial reference. The arm displacement is denoted as x_a and is measured relative to the base. Therefore, the absolute displacement of the arm is

$$y = x_a + x_b. \tag{1}$$

This is the quantity to be compared with the target position y_{target}.

The imposed reference displacement is a chirp signal denoted as x_{ref}. The actuation term can be interpreted as the force generated by the impedance-controlled arm to track the imposed reference:

$$F_a = k_a(x_{ref} - x_a) + r_a(\dot{x}_{ref} - \dot{x}_a), \tag{2}$$

where k_a and r_a are the equivalent stiffness and damping of the actuated arm subsystem.

With the generalized coordinate vector

$$\boldsymbol{q} = \begin{bmatrix} x_a & x_b \end{bmatrix}^T, \tag{3}$$

the Equations of Motion are written as

$$\begin{bmatrix} m_a & m_a \\ m_a & m_a + m_b \end{bmatrix} \begin{bmatrix} \ddot{x}_a \\ \ddot{x}_b \end{bmatrix} + \begin{bmatrix} r_a & 0 \\ 0 & r_b \end{bmatrix} \begin{bmatrix} \dot{x}_a \\ \dot{x}_b \end{bmatrix} + \begin{bmatrix} k_a & 0 \\ 0 & k_b \end{bmatrix} \begin{bmatrix} x_a \\ x_b \end{bmatrix} = \begin{bmatrix} k_a x_{ref} + r_a \dot{x}_{ref} \\ 0 \end{bmatrix}. \tag{4}$$

This formulation is coherent with the choice of x_a as a relative coordinate and x_b as an absolute base coordinate. The corresponding undamped eigenvalue problem used to compute the natural frequencies is

$$\det\left(\boldsymbol{K} - \omega^2 \boldsymbol{M}\right) = 0, \tag{5}$$

with

$$\boldsymbol{M} = \begin{bmatrix} m_a & m_a \\ m_a & m_a + m_b \end{bmatrix}, \qquad \boldsymbol{K} = \begin{bmatrix} k_a & 0 \\ 0 & k_b \end{bmatrix}. \tag{6}$$

1.3 Report Workflow

The report follows the development of the complete optimization and validation pipeline. First, a simulated dataset is generated by sampling mechanical parameters and chirp-command parameters, and by integrating the EoM to obtain the response quantities needed for optimization. The output set of the surrogate model is selected according to the quantities that later enter the BO formulation.

A traditional feed-forward NN is then trained to predict the BO-relevant KPIs. A physics-informed variant is also tested as an intermediate methodological alternative, using a physics-loss term based on the eigenfrequency relation of the undamped system. Since this alternative does not provide a clear advantage for the quantities used by the optimizer, the traditional NN is retained as the final surrogate.

The trained NN is used inside the BO loop to search for a chirp command and arm configuration that satisfy the target-band objective under nominal base conditions. The final BO candidate is then validated with the true EoM, and its behaviour is interpreted through time-domain response, frequency-response and phase diagnostics.

Finally, the robustness of the selected command is assessed through three validation campaigns. The first perturbs the base parameters, the second perturbs the optimized arm parameters, and the third perturbs the initial conditions. A base-specific re-optimization is also performed for one perturbed base configuration, in order to distinguish loss of robustness of the nominal command from actual infeasibility of the perturbed system.

2 Neural network

2.1 Input Space and Dataset Generation

The NN surrogate is designed to approximate a family of base-arm systems rather than a single nominal configuration. For this reason, the dataset generation stage samples both mechanical parameters and chirp-command parameters. The current NN input vector is

$$\boldsymbol{x}_{NN} = [m_b,\ k_b,\ h_b,\ m_a,\ k_a,\ h_a,\ f_{min},\ f_{max},\ T_{chirp},\ A]^T\,. \tag{7}$$

The damping coefficients are computed from the non-dimensional damping ratios as

$$r_b = 2h_b\sqrt{k_b m_b}, \qquad r_a = 2h_a\sqrt{k_a m_a}. \tag{8}$$

A key design choice was to avoid generating the dataset only around the first natural frequency. Although the BO is expected to favor chirps that exploit the first mode, concentrating the entire training set around that region would reduce the generality of the surrogate and could bias the optimizer toward a narrow portion of the dynamic response. The dataset is therefore generated with a Latin Hypercube Sampling strategy over the full prescribed input domain. The final accepted dataset contains 10000 samples.

Table 1: Final input-space definition used for the generation of the NN training dataset.

Input	Meaning	Sampling range	Unit
m_b	Base mass	$[56, 84]$	kg
k_b	Base stiffness	$[3200, 4800]$	N/m
h_b	Base damping ratio	$[0.20, 0.30]$	–
m_a	Arm mass	$[8, 12]$	kg
k_a	Arm stiffness	$[100, 5000]$	N/m
h_a	Arm damping ratio	$[0.20, 1.10]$	–
f_{min}	Chirp lower frequency	$[0.05, 2.00]$	Hz
f_{max}	Chirp upper frequency	$[0.20, 10.00]$	Hz
T_{chirp}	Chirp duration	$[5, 15]$	s
A	Chirp amplitude	$[0.01, 0.10]$	m

Each sample is evaluated by numerically integrating the Equations of Motion introduced in the dynamic-model section. Samples are accepted only if the chirp interval is physically meaningful, namely

$$f_{max} > f_{min} + 0.05\,\mathrm{Hz}, \tag{9}$$

and if the corresponding time-domain simulation is successfully completed. In addition, the generation routine rejects cases producing invalid natural-frequency calculations. No first-mode-centred filtering is imposed at this stage: frequency placement is intentionally left to the BO objective, where the predicted natural frequencies can be used together with the dynamic-response KPIs.

2.2 Output Selection for Bayesian Optimization

The surrogate outputs were selected backwards from the quantities required by the future BO stage. The objective is not to predict every diagnostic quantity that can be extracted from a simulation, but to provide the optimizer with the indicators needed to evaluate reaching performance, frequency placement and feasibility constraints.

The current NN output vector is

$$\boldsymbol{y}_{NN} = \Big[f_{res,1},\ f_{res,2},\ \max_t(y),\ \max_t|\dot{y}|,\ \max_t|F_a|,\ \max_t|x_{ref} - x_a|\Big]^T. \tag{10}$$

These outputs are BO-oriented:

- $f_{res,1}$ and $f_{res,2}$ support the chirp-frequency constraints and the interpretation of the optimized frequency range;
- $\max_t(y)$ is the main reaching-performance indicator;
- $\max_t|\dot{y}|$ is used to check velocity limitations;
- $\max_t|F_a|$ is used to check actuation-force limitations;
- $\max_t|x_{ref} - x_a|$ is used to monitor the internal tracking error between the imposed reference and the relative arm displacement.

2.3 Neural Network Architecture and Training Setup

The selected surrogate is a fully connected feed-forward neural network mapping the 10 input parameters in Eq. (7) to the 6 outputs in Eq. (10). Since the model predicts scalar response indicators rather than time histories, a multilayer perceptron is an appropriate and computationally efficient choice. The adopted architecture is

$$10 \to 128 \to 64 \to 32 \to 6, \tag{11}$$

with ReLU activation functions after each hidden layer. The decreasing hidden-layer width provides enough capacity to learn nonlinear input-output relations while keeping the model compact. This is useful because the surrogate is later called repeatedly inside the BO loop.

The training setup is summarized in Table 2. Input and output normalization are computed using training-set statistics only, and the same scaling is then applied to the validation and test sets. This avoids information leakage from validation or test data into the preprocessing stage.

Table 2: Final training setup of the traditional NN surrogate.

Item	Value
Dataset	10000 accepted samples
Train/validation/test split	70%/15%/15%
Test samples	1500
Architecture	Fully connected MLP, 10-128-64-32-6
Hidden activation	ReLU
Optimizer	Adam
Learning rate	0.002
Batch size	64
Maximum epochs	500
Early stopping monitor	Validation data loss
Early stopping patience	60 epochs
Early stopping min. delta	10^{-5}
Best epoch	453
Best validation data loss	4.013×10^{-3}

2.4 Neural Network Performance

The learning curve in Fig. 3 reports the training and validation data losses over the training epochs. The final model used for testing is the checkpoint corresponding to the best validation loss, not simply the last epoch. This choice is important because it selects the most generalizable model according to the validation set.

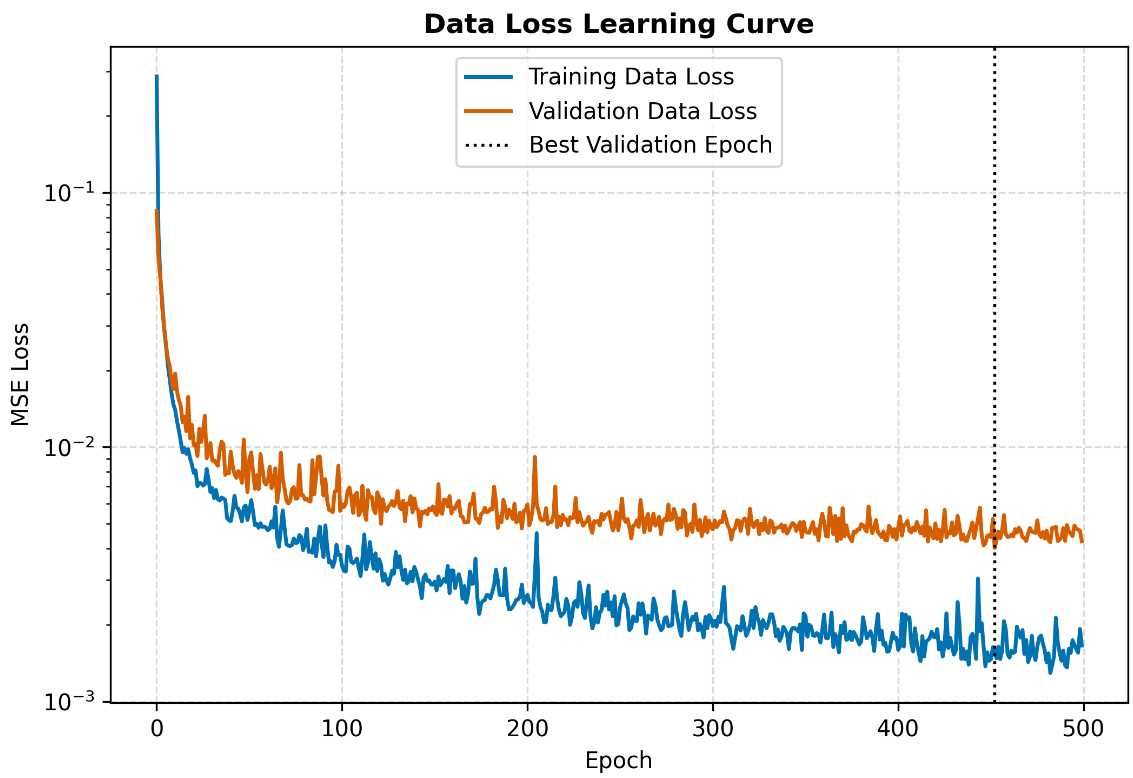

Figure 3: Learning curve of the final traditional NN surrogate. The selected checkpoint corresponds to the best validation data loss.

The two curves decrease rapidly during the first part of the training and then enter a slower refinement phase. The validation loss remains close to the training loss and does not show a sustained upward trend, which suggests that the network is not affected by severe overfitting. The small oscillations visible in the final epochs are consistent with mini-batch training, while the best checkpoint is selected at the epoch with the minimum validation loss.

The following metrics are used to evaluate the NN prediction accuracy:

- R^2: regression-quality indicator comparing the NN predictions with the reference outputs. If the predicted points were plotted against the true values, R^2 close to 1 means that they would lie close to the ideal straight line "prediction = true value".
- MAPE: mean absolute percentage error. It reports the average prediction error as a percentage of the reference value, making outputs with different units easier to compare.
- P95 APE: value below which 95% of the percentage errors fall. For example, a P95 APE of 12% means that 95% of the test samples have an error smaller than approximately 12%. It is useful to detect whether a model has occasional large errors even when the average MAPE is low.

Table 3 reports the test metrics recomputed from the saved best checkpoint. The results indicate high predictive accuracy for both natural frequencies and dynamic-response quantities. The most delicate output remains the maximum actuation force, which has the largest percentage error. This is not unexpected because force peaks are more sensitive to local dynamic effects and damping terms than natural frequencies. Nevertheless, the surrogate accuracy is sufficient for its intended role: guiding the BO toward promising regions of the design space, followed by true-EoM validation of the selected candidates.

Table 3: Test metrics of the final traditional NN surrogate trained with 10000 accepted samples and early stopping.

Output	R^2	**MAPE [%]**	**P95 APE [%]**
$f_{res,1}$	0.9979	0.34	0.91
$f_{res,2}$	0.9993	0.67	1.98
$\max(y)$	0.9851	3.40	12.07
$\max(\lvert\dot{y}\rvert)$	0.9979	3.59	11.81
$\max(\lvert F_a\rvert)$	0.9951	9.18	32.99
$\max(\lvert x_{ref} - x_a\rvert)$	0.9974	3.68	11.76

3 Physics informed NN

3.1 Physics-Loss Formulation

A physics-informed neural network was investigated as an intermediate methodological alternative to the purely data-driven NN. The motivation is that two surrogate outputs, $f_{res,1}$ and $f_{res,2}$, are not arbitrary regression targets: they are constrained by the undamped eigenvalue problem of the two-degree-of-freedom system. Adding a physics residual could therefore improve frequency prediction or make the model less dependent on the available data.

The total loss is written as

$$\mathcal{L}_{tot} = \mathcal{L}_{data} + \lambda_{phys}\mathcal{L}_{phys}, \tag{12}$$

where $\mathcal{L}_{data}$ is the supervised mean squared error on the normalized outputs and λ_{phys} controls the weight assigned to the physics residual. The data term is

$$\mathcal{L}_{data} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{y}}_{NN,sc}^{(i)} - \boldsymbol{y}_{NN,sc}^{(i)}\right\|_2^2, \tag{13}$$

where the subscript *sc* denotes scaled quantities. The physics term penalizes violations of the natural-frequency eigenvalue condition

$$\det\left(\boldsymbol{K} - \omega_j^2\boldsymbol{M}\right) = 0, \qquad j = 1, 2. \tag{14}$$

For each predicted frequency $\hat{f}_{res,j}$, with $\hat{\omega}_j = 2\pi\hat{f}_{res,j}$, the normalized determinant residual is computed as

$$\rho_j = \frac{\det\left(\boldsymbol{K} - \hat{\omega}_j^2\boldsymbol{M}\right)}{k_a k_b + \varepsilon}, \tag{15}$$

and the physics loss is

$$\mathcal{L}_{phys} = \frac{1}{N}\sum_{i=1}^{N}\left[\rho_1^{(i)2} + \rho_2^{(i)2}\right]. \tag{16}$$

The normalization by $k_a k_b$ avoids excessively large determinant scales and makes the residual numerically more stable during training.

This formulation is physically reasonable, but it also relies on an important assumption: the additional physics constraint is applied only to the natural-frequency outputs, while the BO also depends on dynamic outputs such as $\max(y)$, $\max|F_a|$, maximum velocity and tracking error. Therefore, the PINN is not automatically expected to improve the complete surrogate; this has to be checked empirically.

3.2 Selection of the Physics-Loss Weight

Before comparing the traditional NN and the PINN, the value of λ_{phys} was selected through a controlled sweep. This is necessary because the relative scale between $\mathcal{L}_{data}$ and $\mathcal{L}_{phys}$ is not known a priori, and an excessively large physics weight may improve frequency consistency while degrading the BO-relevant dynamic outputs.

The tested values were

$$\lambda_{phys} \in \{0,\ 0.01,\ 0.03,\ 0.05,\ 0.10,\ 0.20,\ 0.50\}. \tag{17}$$

For each value, the same dataset, architecture, preprocessing and split were used. The sweep was repeated over three training seeds to reduce the dependence on random initialization and mini-batch ordering.

The sweep was used as an internal tuning step rather than as a final performance assessment. In particular, aggregate indicators that average errors over frequency outputs and dynamic outputs are not reported here, because such quantities mix physically different targets and can be misleading. The practical conclusion of the sweep was that increasing λ_{phys} did not produce a robust, output-by-output advantage over the data-driven model. A small non-zero value was therefore retained only to define a representative tuned PINN for the subsequent comparison:

$$\lambda_{phys} = 0.03. \tag{18}$$

This value keeps the physics residual active without allowing it to dominate the supervised data loss. The actual comparison with the traditional NN is then performed output by output below.

3.3 Comparison Between Traditional NN and Physics-Informed NN

After selecting a non-zero physics-loss weight, a final controlled comparison was performed between a traditional NN and a tuned physics-informed NN. This comparison is treated as an intermediate surrogate-assessment study, not as the final training run used in the BO. It was carried out on the reduced dataset used for the PINN assessment, consisting of 3000 accepted samples split into 2100 training samples, 450 validation samples and 450 test samples. The final operational NN was instead retrained later on the larger 10000-sample dataset discussed in the Neural network section.

Both compared models used the same architecture, preprocessing, train/validation/test split logic, early-stopping criterion and output vector. The only difference was the value of λ_{phys} in Eq. (12):

$$\text{Traditional NN} : \lambda_{phys} = 0, \qquad \text{Tuned PINN} : \lambda_{phys} = 0.03. \tag{19}$$

The comparison was repeated over five training seeds to reduce the influence of weight initialization, mini-batch ordering and early stopping. Table 4 reports the mean metrics over the five seeds. The color code is assigned according to performance: green denotes the better value within each NN/PINN pair and red denotes the worse value. Therefore, higher is better for R^2, while lower is better for MAPE and P95 APE.

Table 4: Mean comparison over five seeds between traditional NN and tuned physics-informed NN with $\lambda_{phys} = 0.03$. Green indicates the better value within each metric pair.

Output	**NN R^2**	**PINN R^2**	**NN MAPE [%]**	**PINN MAPE [%]**	**NN P95 APE [%]**	**PINN P95 APE [%]**
$f_{res,1}$	0.9706	0.9563	1.06	1.28	3.99	4.51
$f_{res,2}$	0.9985	0.9981	1.07	1.22	3.13	3.92
$\max(y)$	0.9741	0.9735	5.35	5.72	17.79	17.91
$\max(\lvert\dot{y}\rvert)$	0.9932	0.9898	6.02	7.26	19.66	22.87
$\max(\lvert F_a\rvert)$	0.9899	0.9882	11.54	13.47	40.37	46.90
$\max(\lvert x_{ref} - x_a\rvert)$	0.9893	0.9861	6.88	8.62	22.50	25.14

The results are systematic: the traditional NN performs slightly better than the tuned PINN on all the considered outputs. The difference is not dramatic, because both models achieve high R^2 values, but it is consistent in terms of R^2, MAPE and P95 APE. This is relevant because the BO relies not only on the natural frequencies, but also on dynamic indicators such as $\max(y)$, maximum actuation force, maximum velocity and tracking error.

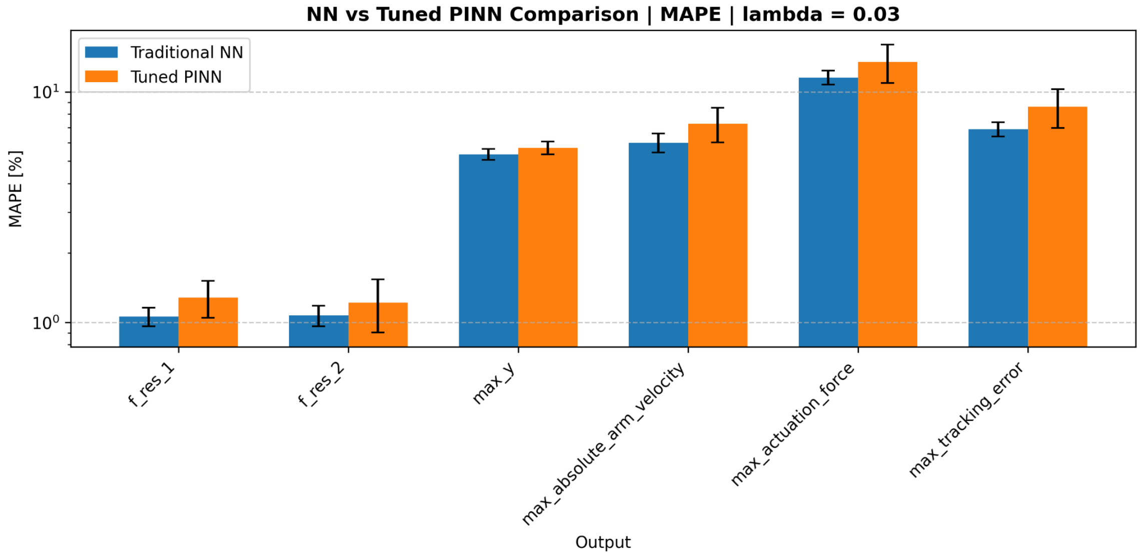

Figure 4: MAPE comparison between traditional NN and tuned PINN over five training seeds. Lower values indicate better predictive accuracy.

Figure 4 confirms the same trend reported in Table 4. The traditional NN achieves lower mean MAPE for all six outputs. The advantage is small for the two natural frequencies, where both models are accurate, but it becomes more relevant for the dynamic quantities used by the BO cost function, especially maximum arm velocity, maximum actuation force and tracking error. The error bars report the variability across the five training seeds, showing that the conclusion is not based on a single random initialization.

The interpretation is that the algebraic physics residual on the natural frequencies does not improve the surrogate in the present formulation. A likely explanation is that the natural frequencies are already directly supervised through the dataset labels. Therefore, the additional residual becomes partly redundant and can interfere with the shared representation used to predict the dynamic KPIs. This does not mean that physics-informed learning is unsuitable for the problem in general. A more structured implementation, such as a multi-head architecture separating frequency prediction from dynamic-response prediction, could still be useful. However, for the current pipeline, the traditional NN is selected as the preferred surrogate for the Bayesian Optimization stage.

4 Bayesian optimization of the chirp command

4.1 BO setup

The final optimization stage selects the mechanical arm parameters and the input-command parameters that maximize the useful dynamic amplification of the compliant base–arm system while respecting the prescribed feasibility requirements. The optimized command is a chirp reference, i.e. a sinusoidal input with time-varying frequency. For the optimized interval, before the final smoothing transient, it is written as

$$x_{ref}(t) = A \sin\left(2\pi f_{min} t + \pi \frac{f_{max} - f_{min}}{T_{chirp}} t^2\right), \qquad 0 \leq t \leq T_{chirp}. \tag{20}$$

This expression makes explicit the four command variables optimized by BO: the lower and upper frequencies f_{min} and f_{max}, the chirp duration T_{chirp}, and the amplitude A.

The Bayesian Optimization (BO) design vector is

$$\boldsymbol{z} = [k_a,\ h_a,\ f_{min},\ f_{max},\ T_{chirp},\ A]^T\,. \tag{21}$$

The base parameters are kept at the nominal values reported in Table 5. These values are the same as the compliant-base validation setup adopted in the reference paper and also correspond to the centre of the base-parameter ranges used to generate the NN training domain. Their effect is analysed later through dedicated robustness and re-optimization studies. The BO search ranges are reported in Table 6; they remain inside the input domain used to train the NN surrogate.

Table 5: Nominal base parameters kept fixed during the BO run.

Parameter	Meaning	Value	Unit
m_b	Base mass	70	kg
k_b	Base stiffness	4000	N/m
h_b	Base damping ratio	0.25	–

Table 6: Design variables optimized by BO.

Variable	Meaning	BO range	Unit
k_a	Arm stiffness	$[100, 5000]$	N/m
h_a	Arm damping ratio	$[0.20, 1.10]$	–
f_{min}	Chirp lower frequency	$[0.05, 2.00]$	Hz
f_{max}	Chirp upper frequency	$[0.20, 10.00]$	Hz
T_{chirp}	Chirp duration	$[5, 15]$	s
A	Chirp amplitude	$[0.01, 0.10]$	m

Table 7: Technical BO setup used in the final target-band optimization run.

Item	Configuration	Role in the implementation
BO design-space representation	Six-dimensional vector mapped to $[0, 1]^6$	Gives comparable numerical scales to all BO variables before fitting the GP.
Objective evaluation inside BO	NN-predicted KPIs inserted into $J(\boldsymbol{z})$	Allows many objective evaluations without integrating the true EoM at each BO step.
Initial sampling	60 Latin-hypercube samples	Builds the initial dataset used to fit the first GP model.
BO iterations	120 sequential acquisitions	Number of adaptive BO updates after the initial design.
GP surrogate for the BO objective	Constant kernel $\times$ Matern kernel ($\nu = 5/2$) + white-noise kernel	Models the scalar objective values observed during BO, including a small regularization/noise term.
GP output treatment	Objective normalization enabled	Improves numerical conditioning when fitting the GP on objective values with different magnitudes.
Acquisition function	Expected Improvement, $\xi = 0.01$	Selects candidates by balancing low predicted objective values and high GP uncertainty.
Acquisition maximization	8000 Latin-hypercube candidate points per iteration	Approximates the maximizer of the acquisition function without solving a separate continuous subproblem.
Post-BO validation	Selected candidates reintegrated with the true EoM	Checks physical admissibility because the final BO search is surrogate-driven.

Table 7 reports the BO configuration rather than the general BO theory. A useful distinction is that two surrogate levels are present in the pipeline. The NN surrogate maps the physical and chirp parameters to the output KPIs used in the cost function, whereas the Gaussian Process is the probabilistic surrogate used internally by BO to model the scalar objective values already observed during the search.

The kernel defines how the GP measures similarity between BO candidates: nearby design points are expected to have similar objective values. A Matern-5/2 kernel was used because it represents a compromise between very smooth kernels, such as the squared-exponential kernel, and rougher Matern alternatives. This is appropriate here because the mechanical response varies continuously with the design variables, while the scalar objective also includes penalties and threshold-based admissibility terms.

Expected Improvement was used as the acquisition function. The parameter $\xi = 0.01$ is a small improvement margin inside the EI criterion: the next point is encouraged to improve the current best objective value by more than an infinitesimal amount. Since this value has to be interpreted with respect to the numerical scale of the BO objective, and since the GP is fitted on normalized objective values, $\xi = 0.01$ represents a mild margin. Much larger values would make the search less sensitive to local refinements and could lead to excessive or poorly informative exploration.

At each iteration, acquisition-function maximization is performed by evaluating Expected

Improvement over 8000 candidate points in the normalized design space and selecting the point with the largest acquisition value for the next objective evaluation. The final acceptability of the selected candidate is nevertheless decided from the true-EoM validation, not from the GP posterior alone.

The target displacement is fixed from the maximum admissible chirp amplitude:

$$y_{target} = 1.5 A_{max} = 0.15\,\mathrm{m}, \tag{22}$$

with $A_{max} = 0.10\,\mathrm{m}$. The target is therefore independent of the candidate value of A. This avoids a moving-target formulation and prevents the optimizer from making the problem artificially easier by reducing the input amplitude. Since $y_{target} > A_{max}$, the target cannot be reached by simply imposing a larger reference displacement; it requires dynamic amplification through the compliant system.

As introduced in the surrogate-model section, the NN is used as a fast evaluator inside the BO loop. In this section, this principle is applied to the final target-band optimization problem: the surrogate guides the search, while the selected candidate is finally validated by re-integrating the true equations of motion (true EoM), especially because the final decision is taken near feasibility thresholds.

4.2 Cost-function formulation

The final BO cost function is designed around a target-band requirement:

$$y_{target} \le \max_t(y) \le y_{target} + M_{allow}, \qquad M_{allow} = 0.01\,\mathrm{m}. \tag{23}$$

In the present case, the admissible interval is therefore

$$0.150\,\mathrm{m} \le \max_t(y) \le 0.160\,\mathrm{m}. \tag{24}$$

This formulation preserves the lower reaching requirement while avoiding candidates that reach the target only through excessive overshoot.

The two target-band penalties are

$$P_{shortfall} = \left(\frac{\max(0, y_{target} - \max_t(y))}{\varepsilon_y} \right)^2, \qquad \varepsilon_y = 0.01\,\mathrm{m}, \tag{25}$$

$$P_{overshoot} = \left(\frac{\max(0, \max_t(y) - y_{target} - M_{allow})}{\varepsilon_{over}} \right)^2, \qquad \varepsilon_{over} = 0.01\,\mathrm{m}. \tag{26}$$

The quantities ε_y and ε_{over} are not numerical-integration tolerances. They are physical normalization scales for the penalties. For instance, with $\varepsilon_y = 0.01\,\mathrm{m}$, a target shortfall of 1 cm gives an unweighted penalty equal to one, while a shortfall of 2 cm gives an unweighted penalty equal to four. The same logic applies to ε_{over}, but only to the overshoot exceeding the allowed margin M_{allow}.

The generic normalized penalty used for upper-limit constraints is

$$P_q = \left[\max\left(0, \frac{q_{max}}{q_{lim}} - 1\right)\right]^2. \tag{27}$$

It is applied to the absolute arm velocity with $v_{lim} = 2\,\mathrm{m/s}$ and to the actuation force with $F_{lim} = 150\,\mathrm{N}$. The force limit is chosen consistently with the force scale discussed in the reference paper for lightweight manipulators, where interaction forces in the range 30 N to 150 N are explicitly considered. The velocity limit is instead introduced as a project-level safety and regularity bound to avoid dynamically aggressive chirp commands.

The chirp interval must also remain feasible:

$$P_{\Delta f} = \left[\max\left(0, \frac{f_{min} + \Delta f_{min} - f_{max}}{\Delta f_{min}}\right)\right]^2, \qquad \Delta f_{min} = 0.05\,\mathrm{Hz}. \tag{28}$$

The minimum gap Δf_{min} only prevents degenerate chirps with f_{max} almost equal to f_{min}. It is small relative to the full frequency domain and therefore does not force the optimizer toward wide-band commands.

A relaxed first-mode containment term is used to encourage the chirp to cover the first natural frequency region:

$$\begin{aligned} P_1 = & \left[\max\left(0, \frac{f_{min} - (1-\eta_1) f_{res,1}}{f_{res,1}}\right)\right]^2 \\ & + \left[\max\left(0, \frac{(1+\eta_1) f_{res,1} - f_{max}}{f_{res,1}}\right)\right]^2, \end{aligned} \tag{29}$$

with $\eta_1 = 0.05$. The 5% margin avoids imposing an unrealistically exact inclusion of the first natural frequency, which is predicted by the surrogate during BO and then rechecked with the true EoM.

The phase check is evaluated at the dominant FRF frequency inside the optimized chirp range:

$$f_{dom} = \arg \max_{f \in [f_{min}, f_{max}]} \left| \frac{Y(j2\pi f)}{X_{ref}(j2\pi f)} \right|. \tag{30}$$

The active phase penalty is

$$P_{\phi,dom} = \left(\frac{\max(0, |\Delta\phi_{y,b}(f_{dom})| - \phi_{lim})}{\phi_{lim}}\right)^2, \qquad \phi_{lim} = 90\,\mathrm{deg}. \tag{31}$$

The value $\phi_{lim} = 90\,\mathrm{deg}$ is an interpretable phase threshold: above this value, the absolute arm response and the base response are no longer predominantly in phase. Since the target is imposed on $y = x_a + x_b$, this check discourages dominant-frequency behaviour in which the base contribution is dynamically unfavourable. It is not used as a formal stability limit, but as a compact dynamic-consistency criterion.

A light chirp-width tie-breaker is finally included as

$$P_{band} = \left(\frac{f_{max} - f_{min}}{f_{max}^{UB} - f_{min}^{LB}}\right)^2, \tag{32}$$

where $f_{max}^{UB} = 10\,\text{Hz}$ and $f_{min}^{LB} = 0.05\,\text{Hz}$ are the global bounds of the BO frequency domain. This term acts as a weak regularization on the chirp bandwidth $\Delta f = f_{\max} - f_{\min}$. It does not represent a primary physical constraint, but discourages unnecessarily wide frequency sweeps. The objective is to allow a chirp bandwidth large enough to excite the relevant dynamics, while avoiding solutions where the optimizer simply selects a very broad sweep because it will eventually cross a useful frequency region. In this way, the optimized command is encouraged to remain focused on the dynamically relevant frequency interval rather than relying on excessive frequency coverage. Since its weight is much smaller than the weights associated with target reaching and constraint satisfaction, this contribution mainly acts as a tie-breaker among otherwise comparable candidates.

The final active cost function is

$$\begin{aligned} J = 20P_{shortfall} + 20P_{overshoot} + 20P_{\Delta f} + 0.5P_1 \\ + P_{\phi,dom} + 10P_v + 0.2P_F + 0.04P_{band}. \end{aligned} \tag{33}$$

The active terms are summarized in Table 8. The target-band and chirp-validity terms are dominant because they define the primary feasibility requirements. The first-mode and phase terms are kept weaker because they guide the optimizer toward physically meaningful amplification without making the frequency-domain interpretation too rigid. The velocity weight is relatively high because velocity is directly related to the aggressiveness of the chirp and to the damping contribution in the actuation force; the normalized formulation ensures that only actual relative violations are penalized. Conversely, the force term is kept soft because an excessively strong force penalty can drive the optimizer toward conservative candidates that fail to reach the prescribed displacement band. The maximum tracking error is predicted and monitored as a diagnostic quantity, but it is not used as a final hard requirement in the BO objective.

Table 8: Active BO cost-function terms used in the final target-band formulation.

Term	Weight	Role
$P_{shortfall}$	20	Primary lower-bound target-band requirement
$P_{overshoot}$	20	Primary upper-bound target-band requirement
$P_{\Delta f}$	20	Reject invalid or nearly degenerate chirp ranges
P_1	0.5	Encourage inclusion of the first natural-frequency region
$P_{\phi,dom}$	1	Avoid unfavourable dominant-frequency base-arm phase behaviour
P_v	10	Penalize excessive absolute arm velocity above 2 m/s
P_F	0.2	Soft force-aware term with $F_{lim} = 150\,\text{N}$
P_{band}	0.04	Light tie-breaker on chirp width

4.3 Final BO result

After the optimized chirp, a short smooth fade-out is appended to avoid an abrupt interruption of the arm reference motion. This final transient is not part of the BO design vector and is not used to artificially improve target reaching; its role is only to make the end of the command dynamically smoother during true-EoM validation.

The final BO run selects the candidate reported in Table 9. The result is validated by re-integrating the true EoM with the smooth fade-out included.

Table 9: Final BO candidate selected for the nominal base configuration.

Candidate	k_a [N/m]	h_a [-]	f_{min} [Hz]	f_{max} [Hz]	T_{chirp} [s]	A [m]	max(y) [m]	v_{max} [m/s]	F_{max} [N]	$\Delta\phi_{y,b}(f_{dom})$ [deg]	J_{true}
Target-band BO	289.02	0.5085	0.436	1.347	12.664	0.09976	0.15060	0.778	48.43	24.65	3.35×10^{-4}

The candidate satisfies the target band with a small positive reaching margin:

$$\max(y)_{EoM} - y_{target} = 0.150\,604\,\mathrm{m} - 0.150\,000\,\mathrm{m} = 0.000\,604\,\mathrm{m}. \tag{34}$$

The remaining margin to the upper bound is

$$0.160\,000\,\mathrm{m} - 0.150\,604\,\mathrm{m} = 0.009\,396\,\mathrm{m}. \tag{35}$$

Therefore, the nominal solution reaches the prescribed target band with high precision. At the same time, the reaching margin is small, so robustness cannot be inferred from the nominal case alone and must be assessed explicitly in the following sections. The velocity and force values are also below their monitored limits:

$$v_{max} = 0.778\,\mathrm{m/s} < 2\,\mathrm{m/s}, \qquad F_{max} = 48.43\,\mathrm{N} < 150\,\mathrm{N}. \tag{36}$$

The same candidate is compared with the NN prediction in Table 10. The surrogate is accurate enough to guide the BO toward a useful region of the design space, but it is not accurate enough to replace final true-EoM validation. In particular, the NN slightly overestimates the peak displacement: it predicts $\max(y) = 0.153\,858\,\mathrm{m}$, whereas the true EoM gives $\max(y) = 0.150\,604\,\mathrm{m}$. This difference is acceptable because the true response still lies inside the target band, but it confirms that EoM validation is decisive near target thresholds.

Table 10: NN-surrogate prediction versus true-EoM validation for the final candidate on the nominal base.

Quantity	**NN prediction**	**True EoM**	**Relative error [%]**
$f_{res,1}$ [Hz]	0.7858	0.8046	2.34
$f_{res,2}$ [Hz]	1.3018	1.2793	1.76
max(y) [m]	0.153858	0.150604	2.16
v_{max} [m/s]	0.8280	0.7781	6.41
F_{max} [N]	71.47	48.43	47.59
Maximum tracking error [m]	0.1117	0.1052	6.20

Additional diagnostics are reported in Table 11. The table is intentionally limited to quantities that add information beyond the optimized design variables and the main true-EoM validation metrics already reported above.

Table 11: Diagnostics for the final BO candidate.

Diagnostic	**Value**	**Meaning**
G_2/G_1	0.876	Ratio between the second and first FRF amplification peaks; values below one indicate that the first peak remains dominant.
Dominant frequency in chirp f_{dom}	0.715 Hz	Frequency inside the optimized chirp band where the absolute-displacement FRF reaches its largest value.
$\Delta\phi_{y,b}(f_{dom})$	24.65 deg	Absolute arm-minus-base phase at the dominant frequency; it checks whether the base contribution is dynamically favourable where the response is strongest.
Final-window peak-to-peak displacement	2.38×10^{-4} m	Residual oscillation amplitude in the final simulation window; it verifies that the system has almost settled after the command and fade-out.

As a consistency check between the frequency-domain interpretation and the time-domain validation, the effective amplification obtained with the optimized chirp is

$$\frac{\max(y)}{A} = \frac{0.150604}{0.099756} = 1.510. \tag{37}$$

The maximum FRF gain inside the optimized chirp range is $G(f_{dom}) = 1.541$. The difference between these two values is approximately 2%, which supports the use of the FRF as a meaningful diagnostic for this candidate. This comparison should not be interpreted as an exact equality, because the chirp response remains transient, but it indicates that the selected sweep is not too fast to exploit the dominant amplification region.

Figure 5 reports the surrogate-based BO convergence. The plot documents how the optimizer progressively identifies low-objective candidates, but it should not be interpreted alone as proof of physical validity because the optimization objective is evaluated on the NN surrogate. The decisive check is the true-EoM validation.

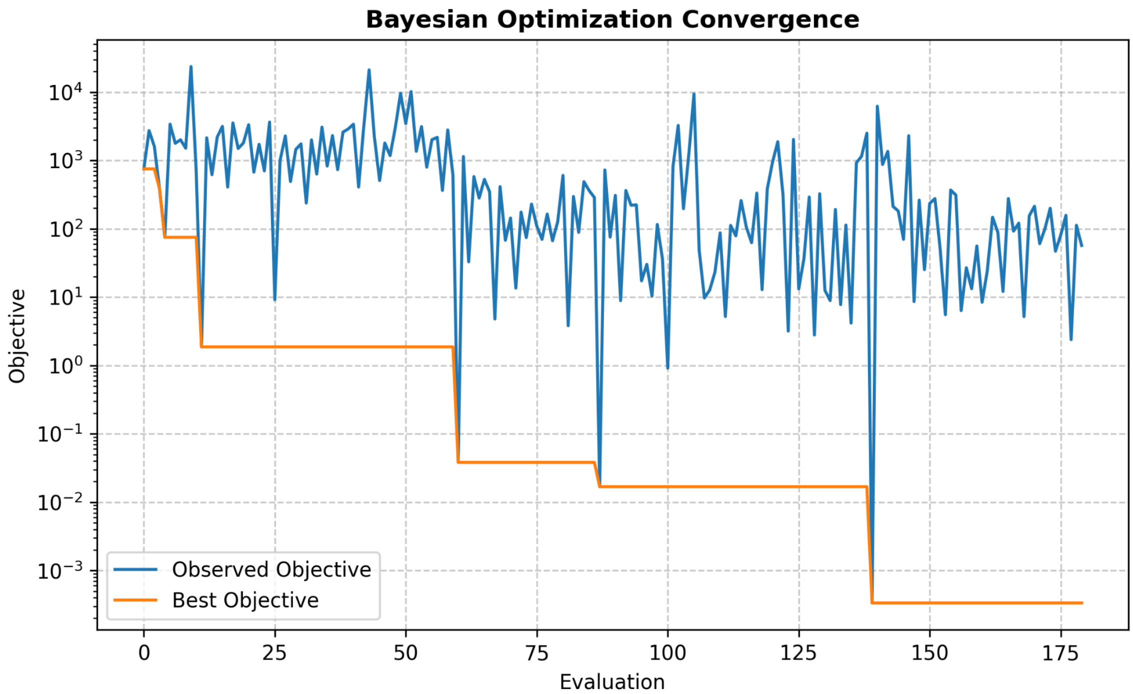

Figure 5: Bayesian Optimization convergence for the final target-band run.

The validated time-domain response is shown in Fig. 6. The peak absolute displacement lies just above the lower target bound and below the allowed overshoot bound. The same plot also confirms that the velocity and force constraints are not active for the selected nominal candidate.

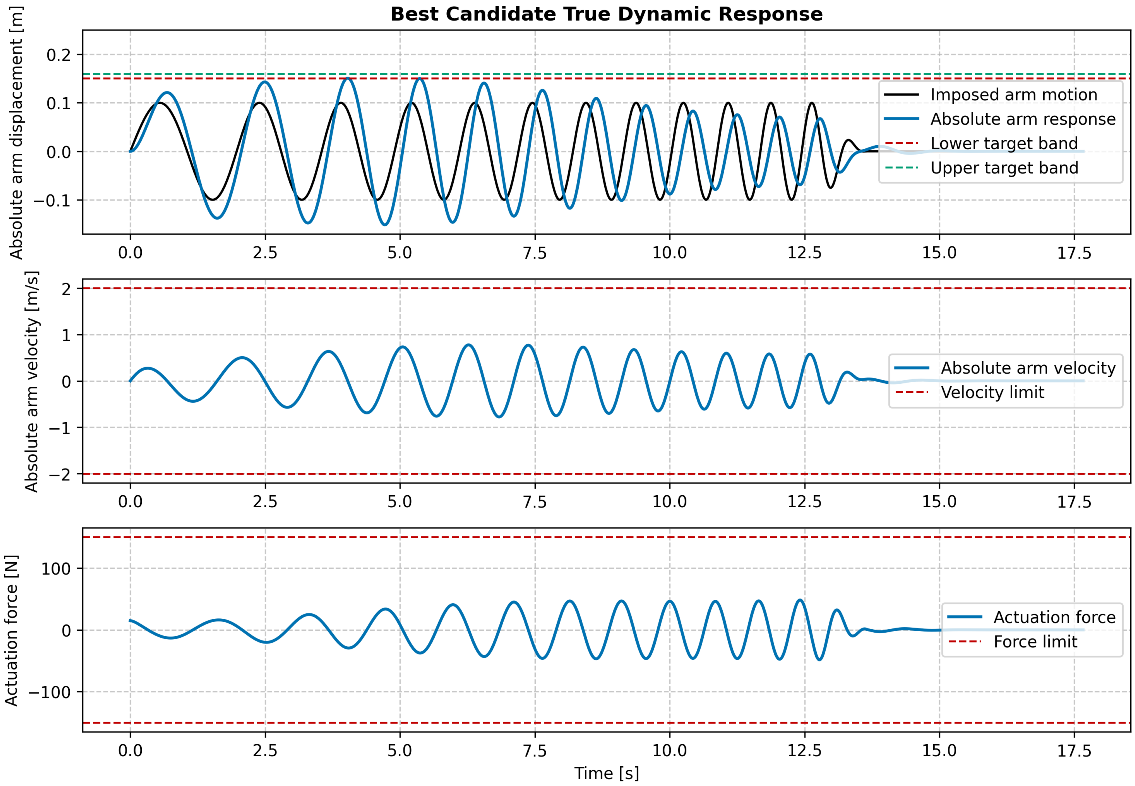

Figure 6: True-EoM time-domain response of the final BO candidate.

The FRF in Fig. 7 is used as a frequency-domain diagnostic to interpret why the optimized chirp is effective. Since a chirp is a transient excitation, the FRF should not be read as a direct prediction of the exact time-domain peak. Nevertheless, the comparison is meaningful if the chirp sweep is not excessively fast: the system must spend enough time near the relevant resonant region to develop dynamic amplification. This is one reason why T_{chirp} is included among the BO variables instead of being fixed a priori.

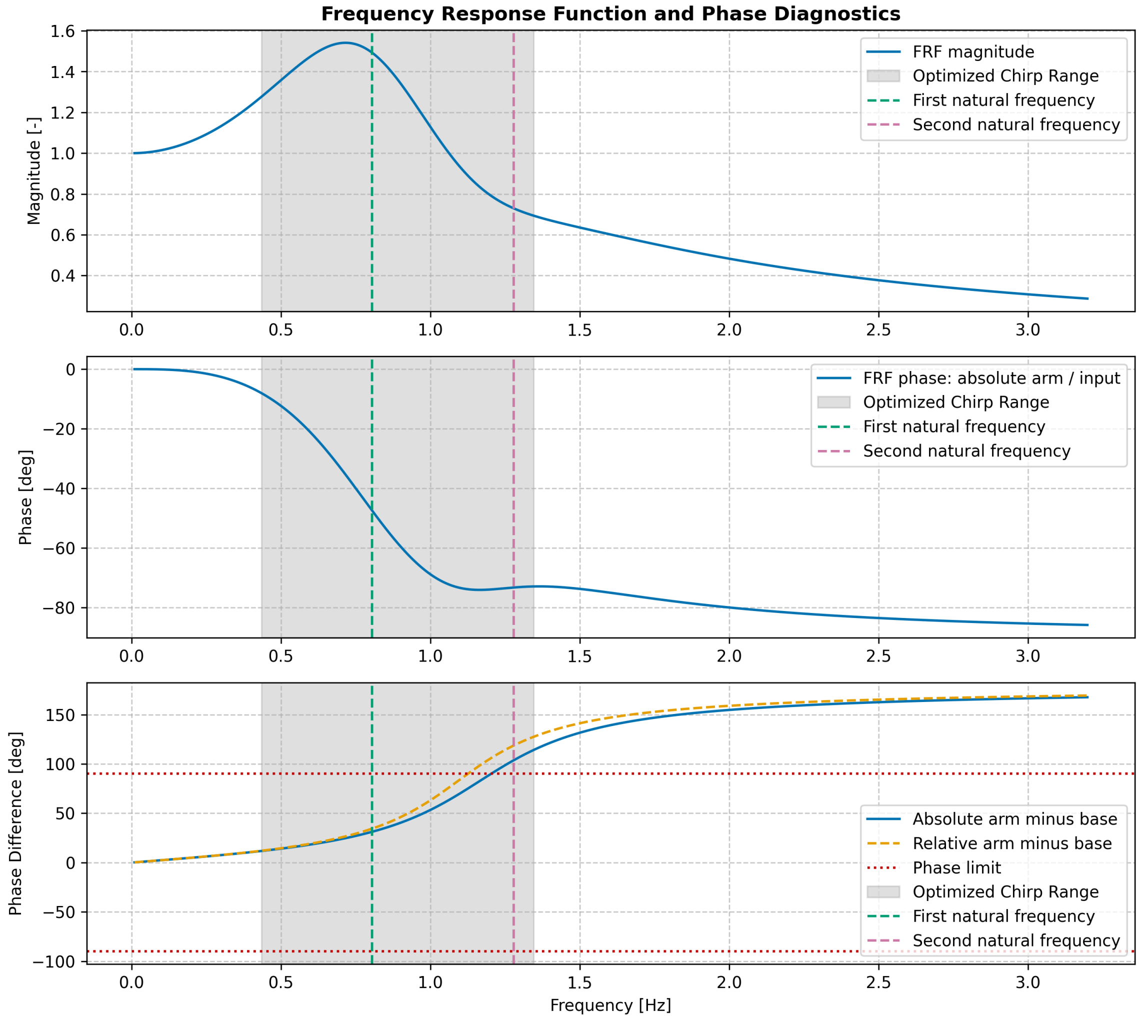

Figure 7: Frequency response function and phase diagnostics for the final BO candidate.

Figure 8 reports two phase indicators over the optimized chirp range. The relative arm-minus-base phase compares the arm deformation relative to the base with the base motion. The absolute arm-minus-base phase compares the absolute arm displacement $y = x_a + x_b$ with the base motion. The latter is more directly related to the BO target because the cost function is defined on the absolute displacement y. At the dominant response frequency, the absolute arm-minus-base phase remains well below the adopted 90 deg threshold.

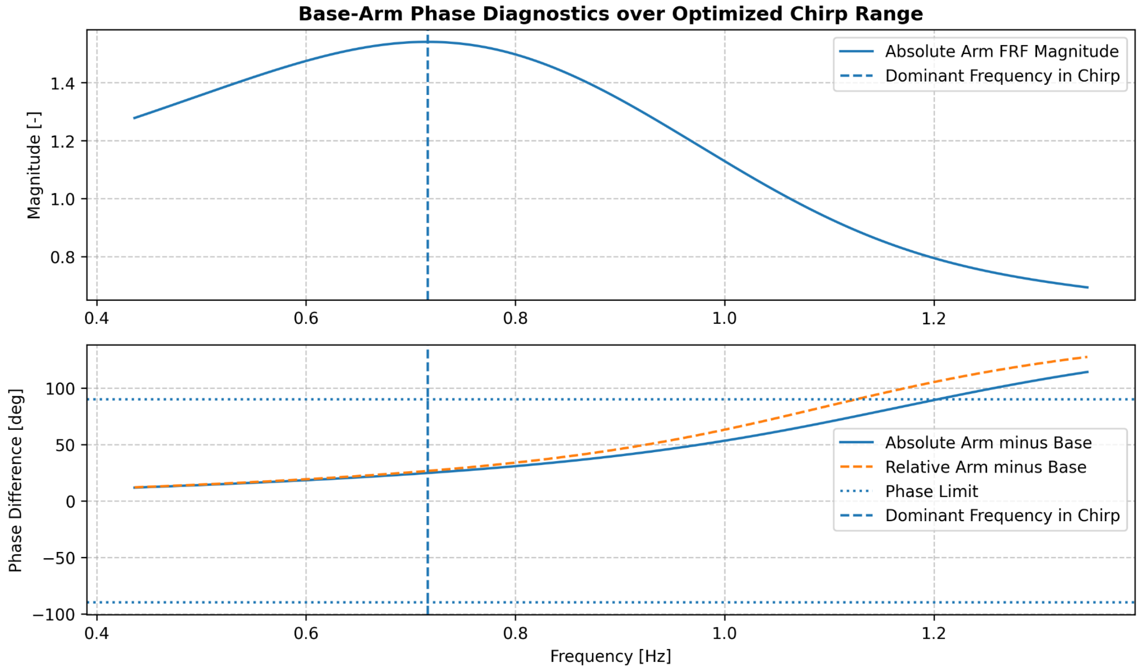

Figure 8: Base-arm phase diagnostics over the optimized chirp range.

5 Robustness validation: base parameters perturbation

After selecting the final v11 target-band candidate, a dedicated robustness campaign was performed to assess how the fixed nominal command behaves when the compliant-base parameters are perturbed. The optimized arm and chirp parameters were kept fixed at the final BO values reported in Table 9, while only the base parameters were changed. The purpose of this campaign is not to prove global robustness over an arbitrary physical domain, but to identify the region in which the already optimized command remains admissible without re-running the Bayesian Optimization.

The robustness envelope is intentionally broad and should be interpreted as a stress test, not as a statistical uncertainty model of the base. In other words, the tested ranges are used to expose failure mechanisms and validity boundaries of the nominal command, rather than to represent a probability distribution of manufacturing or identification errors.

5.1 Stress-test definition

The nominal base configuration is the one defined in the BO setup, Table 5. In this section the perturbed quantities are therefore expressed through the non-dimensional ratios

$$\frac{m_b}{m_{b,0}}, \qquad \frac{k_b}{k_{b,0}}, \qquad \frac{h_b}{h_{b,0}}. \tag{38}$$

The mass and stiffness ratios were varied over the range

$$\frac{m_b}{m_{b,0}} \in [0.25, 1.75], \qquad \frac{k_b}{k_{b,0}} \in [0.25, 1.75], \tag{39}$$

using 13 equally spaced levels for each parameter. The damping-ratio perturbation was tested at five levels:

$$\frac{h_b}{h_{b,0}} \in \{0.25,\ 0.625,\ 1.00,\ 1.375,\ 1.75\}. \tag{40}$$

The complete campaign therefore contains

$$13 \times 13 \times 5 = 845 \tag{41}$$

base configurations.

All robustness cases were evaluated using the true EoM, not the NN surrogate. This choice is important because this stage is a validation step rather than a fast design-space exploration step. During BO, the surrogate is useful to reduce the cost of proposing candidate commands; during robustness validation, the objective is instead to compute accurate margins for one already selected command. This is particularly relevant because several tested base configurations are far from the nominal point used in the final BO run.

5.2 Admissibility criteria and signed margins

The robustness analysis is based on signed margins. The target-band response is evaluated through the reaching margin

$$M_y = \max_t(y) - y_{target}. \tag{42}$$

This quantity measures the signed distance between the peak absolute displacement and the lower target bound. If $M_y < 0$, the response presents a target shortfall. If $0 \le M_y \le M_{allow}$, the target band is satisfied. If $M_y > M_{allow}$, the lower target is reached but the admissible upper target-band limit is exceeded, resulting in target overshoot.

The velocity, force and dominant-phase margins are defined as

$$M_v = v_{lim} - v_{max}, \qquad M_F = F_{lim} - F_{max}, \qquad M_\phi = \phi_{lim} - |\Delta\phi_{y,b}(f_{dom})|. \tag{43}$$

For these three quantities, a positive value means that the corresponding limit is respected, while a negative value means that the limit is exceeded.

A distinction is made between a numerically successful simulation and an admissible case. A simulation is numerically successful if the time-domain integration is completed. A case is instead classified as admissible only if the target-band condition and all active safety margins are satisfied:

$$\begin{aligned} &0 \le M_y \le M_{allow}, \\ &M_v \ge 0, \qquad M_F \ge 0, \qquad M_\phi \ge 0. \end{aligned} \tag{44}$$

This definition is stricter than lower-target reaching alone, because excessive overshoot is explicitly treated as a failure mode in the final target-band formulation.

5.3 Base-parameter robustness results

All 845 integrations were numerically successful. Among them, 337 cases satisfied all admissibility criteria, corresponding to 39.9% of the grid. The dominant failure mechanism is target shortfall: 458 cases, or 54.2% of the tested configurations, do not reach y_{target}. A smaller set of 50 cases, or 5.9%, exceeds the upper target-band limit. This confirms that both sides of the target band are relevant: the nominal v11 command is precise at the nominal base configuration, but base perturbations can either reduce the amplification below the target or amplify the response excessively.

The minimum reaching margin is

$$M_{y,min} = -0.032\,43\,\mathrm{m}, \tag{45}$$

obtained at

$$\frac{m_b}{m_{b,0}} = 1.375, \qquad \frac{k_b}{k_{b,0}} = 0.25, \qquad \frac{h_b}{h_{b,0}} = 0.625. \tag{46}$$

The largest positive reaching margin is instead

$$M_{y,max} = 0.043\,28\,\mathrm{m}, \tag{47}$$

which corresponds to an excessive target overshoot case at

$$\frac{m_b}{m_{b,0}} = 0.25, \qquad \frac{k_b}{k_{b,0}} = 0.25, \qquad \frac{h_b}{h_{b,0}} = 0.25. \tag{48}$$

Thus, a positive reaching margin is not sufficient by itself: if it exceeds $M_{allow} = 0.01\,\mathrm{m}$, the upper target-band requirement is violated.

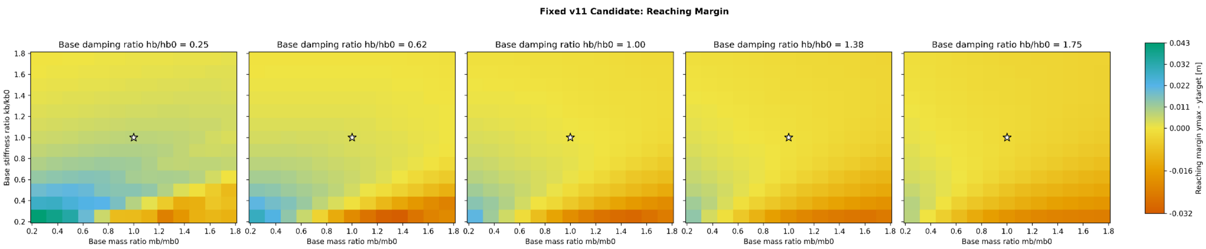

Figure 9: Reaching margin of the fixed v11 command over the base-parameter stress-test envelope. Positive values indicate that the lower target is reached, but excessively positive values must also be checked against the upper target-band limit.

The reaching-margin heatmap in Fig. 9 shows that the margin generally decreases when the base mass and damping increase. This trend is confirmed by the Spearman coefficients $\rho(M_y, m_b/m_{b,0}) = -0.51$ and $\rho(M_y, h_b/h_{b,0}) = -0.57$. Physically, heavier or more damped bases reduce the dynamic amplification that the optimized chirp was designed to exploit. The global stiffness correlation is weak, $\rho(M_y, k_b/k_{b,0}) = +0.04$, meaning that stiffness affects the response mainly through coupled interactions with mass and damping rather than through a simple monotonic trend over the whole grid.

The velocity limit is never active in the base stress test. The maximum observed velocity is

$$v_{max,max} = 0.948\,\mathrm{m/s}, \tag{49}$$

well below $v_{lim} = 2\,\mathrm{m/s}$. Nevertheless, the velocity-margin heatmap remains informative because it shows how the dynamic aggressiveness of the response changes across the base domain.

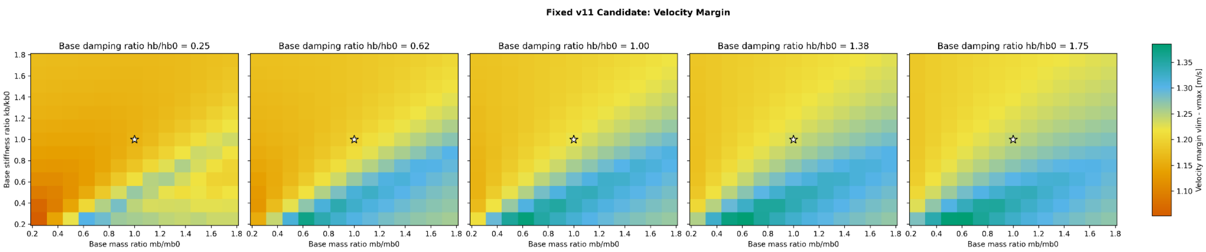

Figure 10: Velocity margin over the base stress-test envelope. The velocity limit is respected in all tested configurations.

Since $M_v = v_{lim} - v_{max}$, a larger velocity margin means a less critical velocity response. The Spearman coefficients indicate that the velocity margin tends to increase with mass and damping, $\rho = +0.51$ and $\rho = +0.46$, while it decreases with stiffness, $\rho = -0.54$. Thus, heavier and more damped bases tend to reduce the peak velocity, whereas stiffer bases tend to produce higher velocities. However, the available margin remains large in all cases.

The force limit is also never active. The maximum observed actuation force is

$$F_{max,max} = 62.91\,\mathrm{N}, \tag{50}$$

which is well below $F_{lim} = 150\,\mathrm{N}$.

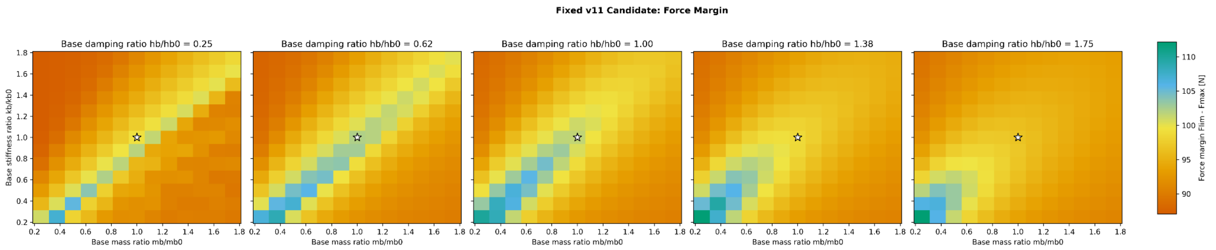

Figure 11: Force margin over the base stress-test envelope. All tested configurations remain below the adopted 150 N threshold.

The force margin is much less sensitive to the base perturbations than the reaching and phase margins. The correlations are weak: $\rho(M_F, m_b/m_{b,0}) = -0.05$, $\rho(M_F, k_b/k_{b,0}) = -0.15$ and $\rho(M_F, h_b/h_{b,0}) = +0.12$. This confirms that the main limitation of the fixed nominal command is not force saturation, but the target-band sensitivity caused by the change in base dynamics.

The dominant phase criterion is not a primary limitation of the base robustness campaign. Only 5 cases violate the dominant 90 deg phase threshold, corresponding to 0.6% of the grid. The maximum absolute dominant phase difference is 165.1 deg; however, these phase-critical configurations are already located in the target-shortfall region. Therefore, dominant phase degradation should be interpreted as a secondary symptom of strongly shifted base dynamics rather than as the main independent failure mode.

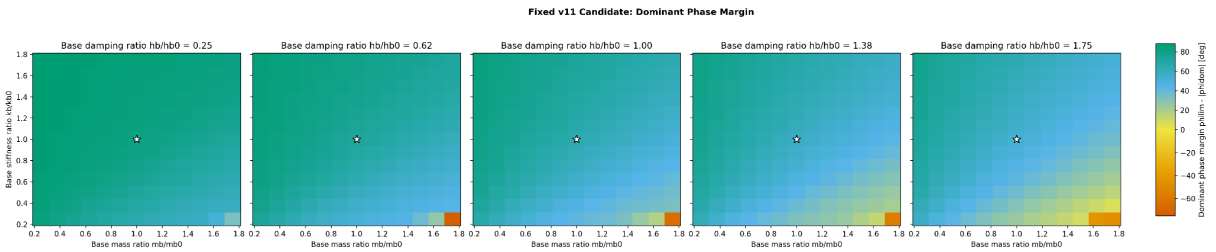

Figure 12: Dominant base-arm phase margin over the base stress-test envelope. The phase criterion is violated only in a very small region and is not the dominant failure mode.

The phase margin decreases with increasing base mass and damping, with $\rho(M_\phi, m_b/m_{b,0}) = -0.53$ and $\rho(M_\phi, h_b/h_{b,0}) = -0.64$. Conversely, increasing base stiffness tends to improve the phase margin, with $\rho(M_\phi, k_b/k_{b,0}) = +0.50$. This is consistent with the interpretation that heavier and more damped bases shift the dynamic contribution of the base away from the nominal amplification mechanism.

The category map in Fig. 13 gives the clearest summary of the admissible domain. It separates fully admissible cases from target shortfall and target overshoot cases. Increasing base damping and/or mass generally reduces amplification and leads to target shortfall. Conversely, very light and weakly damped configurations may amplify the response too much and violate the upper target-band limit.

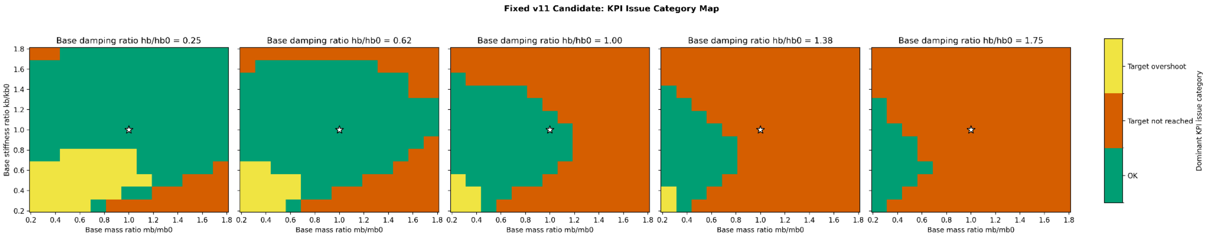

Figure 13: Dominant KPI issue category for the fixed v11 command under base-parameter perturbations. Green regions identify fully admissible cases, while orange and yellow regions correspond to target shortfall and excessive target overshoot, respectively.

The violation summary in Fig. 14 confirms that target-band satisfaction is the active limitation. Velocity and force violations are absent, target shortfall is dominant, and target overshoot appears as a secondary issue.

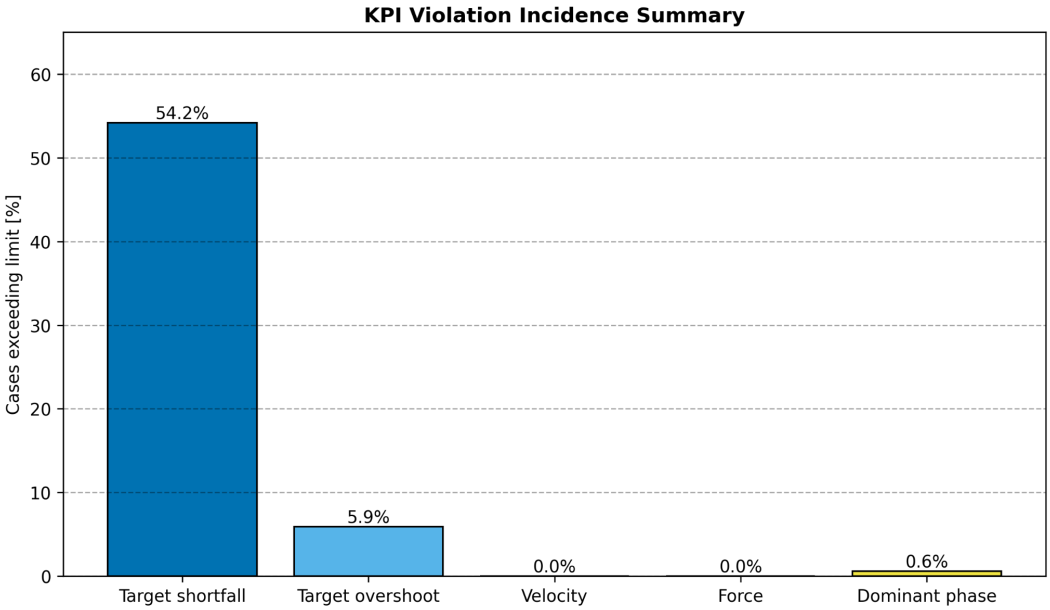

Figure 14: Percentage of base stress-test simulations violating each monitored criterion. Target shortfall is the dominant failure mode, followed by excessive target overshoot.

Table 12: Summary of the fixed v11 base-parameter robustness campaign.

Criterion	Cases	Fraction [%]
Numerically successful simulations	845	100.0
Target band satisfied / admissible	337	39.9
Target shortfall	458	54.2
Target overshoot	50	5.9
Velocity limit exceeded	0	0.0
Force limit exceeded	0	0.0
Dominant phase limit exceeded	5	0.6

5.4 FRF and sensitivity interpretation

A one-at-a-time FRF analysis was performed by perturbing mass, stiffness and damping separately while keeping the other base parameters nominal. The fixed v11 chirp window is

$$[f_{min}, f_{max}] = [0.436\,\text{Hz}, 1.347\,\text{Hz}]. \tag{51}$$

The purpose of this plot is not only to check whether a modal peak remains inside the chirp interval. It also helps distinguish two different effects: a frequency shift of the modal peaks and a change in the amplification quality of the response. A peak may remain inside the chirp window while the effective target response still changes because the modal gain, damping and phase relations have been modified.

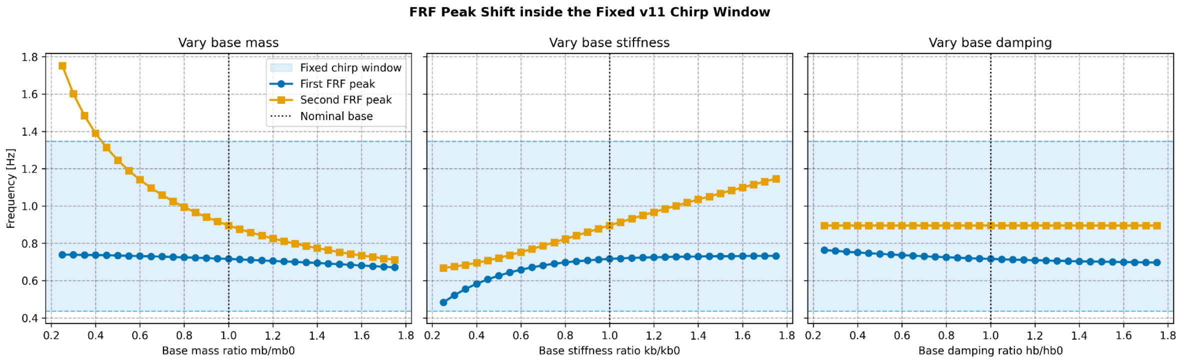

Figure 15: One-at-a-time displacement of the first and second FRF peaks relative to the fixed v11 chirp window. The first peak remains inside the excitation interval, while the second peak can move outside the upper bound for very light bases.

Figure 15 shows that the first FRF peak remains inside the optimized chirp window throughout the one-at-a-time tests. Therefore, the nominal command does not completely lose contact with the main amplification region. However, the robustness results show that this is not sufficient to guarantee admissibility over the full stress grid: variations in mass and damping can still reduce the useful amplification and generate target shortfall, while very light and weakly damped bases can produce excessive amplification. The second peak remains inside the chirp window for nominal and high-mass cases, but it can move above the upper chirp frequency when the base mass is strongly reduced.

To summarize the global monotonic trends, the Spearman rank-correlation coefficient was computed over the complete grid. Spearman correlation is used because it measures monotonic association without assuming a linear relationship. This is appropriate here because the response to base-parameter perturbations is coupled and is not expected to be globally linear.

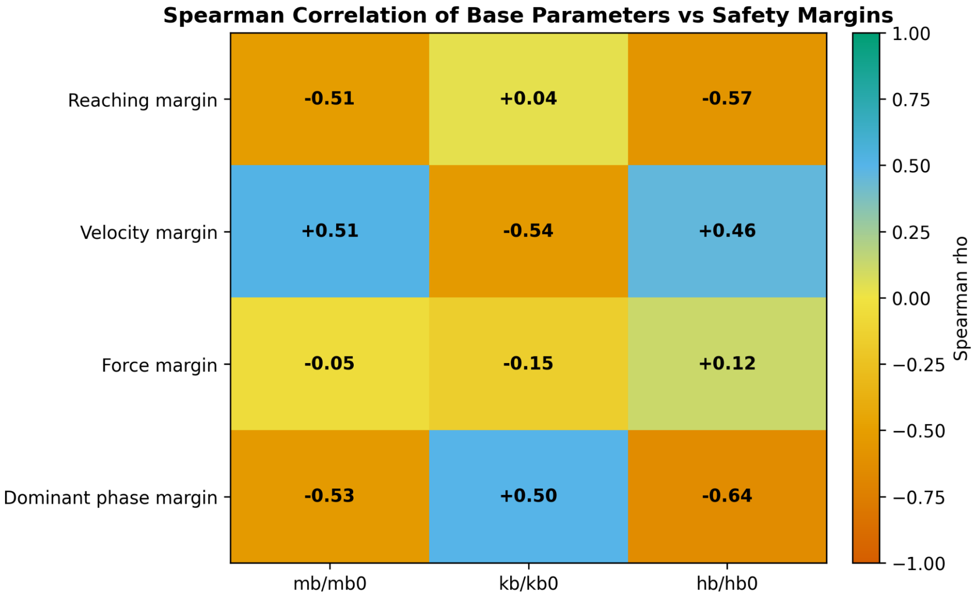

Figure 16: Spearman correlation between the perturbed base parameters and the safety margins. Negative values indicate that increasing the parameter tends to reduce the corresponding margin.

The Spearman map in Fig. 16 confirms the main trends already visible in the heatmaps. Reaching and phase margins are most strongly reduced by increasing base mass and damping. Velocity margin follows the opposite trend with mass and damping because the peak velocity becomes smaller. The force margin remains weakly correlated with all base parameters, confirming that force is not the active robustness limitation in this campaign.

5.5 Verified base validity domain

The stress grid should not be interpreted only through the global admissible percentage. A more useful outcome is the verified validity domain of the fixed v11 command. With the other two base parameters kept nominal, the one-at-a-time admissible ranges are

$$\frac{m_b}{m_{b,0}} \in [0.25, 1.125], \qquad \frac{k_b}{k_{b,0}} \in [0.625, 1.125], \qquad \frac{h_b}{h_{b,0}} \in [0.25, 1.00]. \tag{52}$$

These are one-at-a-time ranges and should not be confused with a simultaneous validity box.

A conservative simultaneous all-admissible box identified directly from the tested grid is

$$\boxed{\frac{m_b}{m_{b,0}} \in [0.25, 1.00], \qquad \frac{k_b}{k_{b,0}} \in [0.875, 1.125], \qquad \frac{h_b}{h_{b,0}} \in [0.25, 1.00]}. \tag{53}$$

All tested combinations inside this box are admissible. This box is discrete and conservative: it is derived from the tested grid points and should not be interpreted as a continuous mathematical guarantee between all sampled points. Its asymmetry is physically meaningful. The v11 solution tolerates reductions in base mass and damping better than increases in these parameters, because increasing mass or damping reduces the dynamic amplification required to reach the target band.

This result also motivates the base-specific re-optimization study reported below. The present robustness map shows where the fixed nominal v11 command loses admissibility when the base changes. It does not imply that those perturbed systems are infeasible: a failed nominal command may simply require a different chirp and arm-parameter combination.

5.6 Base-specific re-optimization on a perturbed base

Only one base-specific re-optimization was retained in the final report. It corresponds to the v12 BO run introduced after the v11 base-robustness campaign. The goal is not to perform a new robustness sweep, but to test whether one base configuration that is not admissible with the nominal v11 command can still be made admissible by re-running the same BO pipeline for that specific physical system.

The selected perturbed base is

$$\frac{m_b}{m_{b,0}} = 1.125, \qquad \frac{k_b}{k_{b,0}} = 1.125, \qquad \frac{h_b}{h_{b,0}} = 1.000, \tag{54}$$

corresponding to

$$m_b = 78.75\,\mathrm{kg}, \qquad k_b = 4500\,\mathrm{N/m}, \qquad h_b = 0.25. \tag{55}$$

This case is useful because it is a moderate perturbation, remains inside the training envelope of the NN surrogate, and yet the nominal v11 candidate already fails by a small target shortfall. Therefore, the comparison separates two different questions: the nominal command is not robust to this base perturbation, but the perturbed system may still be controllable after re-optimizing the arm and chirp parameters.

No change was made to the BO budget or to the cost-function structure. The same NN surrogate, design bounds, target-band formulation, random seed, number of initial samples and number of BO iterations used for the nominal v11 run were retained. The only physical change inside the BO was the fixed base configuration. The comparison is intentionally limited to the same perturbed base: the nominal v11 candidate is evaluated on the perturbed base, and the new base-specific candidate is evaluated on the same perturbed base.

The resulting design is compared with the nominal v11 candidate in Table 13. The arm stiffness and command amplitude remain close to the nominal-v11 values, while the damping ratio and the chirp range change substantially. The base-specific solution reduces the equivalent arm damping and uses a higher, wider and shorter chirp. This indicates that the recovery of target reaching is mainly obtained by changing the dynamic excitation strategy rather than by simply increasing the imposed amplitude.

Table 13: Design comparison between the nominal v11 candidate and the base-specific BO candidate. Both are evaluated on the same perturbed base.

Candidate	k_a [N/m]	h_a [-]	f_{min} [Hz]	f_{max} [Hz]	T_{chirp} [s]	A [m]
Nominal BO v11 candidate	289.018	0.5085	0.4361	1.3466	12.6639	0.099756
Base-specific BO candidate	293.555	0.2848	0.6987	2.8257	8.5254	0.099480

The true-EoM comparison is reported in Table 14. The nominal v11 candidate gives $\max(y) = 0.149\,826\,\mathrm{m}$ on the perturbed base, corresponding to a reaching margin of $-0.174\,\mathrm{mm}$ and therefore to a small target shortfall. The base-specific candidate instead reaches $\max(y) = 0.157\,711\,\mathrm{m}$, corresponding to $M_y = 7.711\,\mathrm{mm}$, which lies inside the admissible band $0 \leq M_y \leq 10\,\mathrm{mm}$. The velocity, force and dominant-phase constraints are also respected.

Table 14: True-EoM comparison on the selected perturbed base. The re-optimized candidate recovers the target band without violating the velocity, force or dominant-phase constraints.

Candidate	max(y) [m]	M_y [mm]	Target-band outcome	v_{max} [m/s]	F_{max} [N]	Admissible
Nominal BO v11 on perturbed base	0.149826	-0.174	Shortfall	0.779	49.58	No
Base-specific BO on perturbed base	0.157711	7.711	Inside band	0.945	67.01	Yes

The time-domain comparison in Fig. 17 gives the same interpretation visually. The nominal v11 command remains mechanically mild, but its displacement peak stays just below the lower target bound. The base-specific command produces a larger response peak inside the target band, while remaining well below the velocity and force limits.

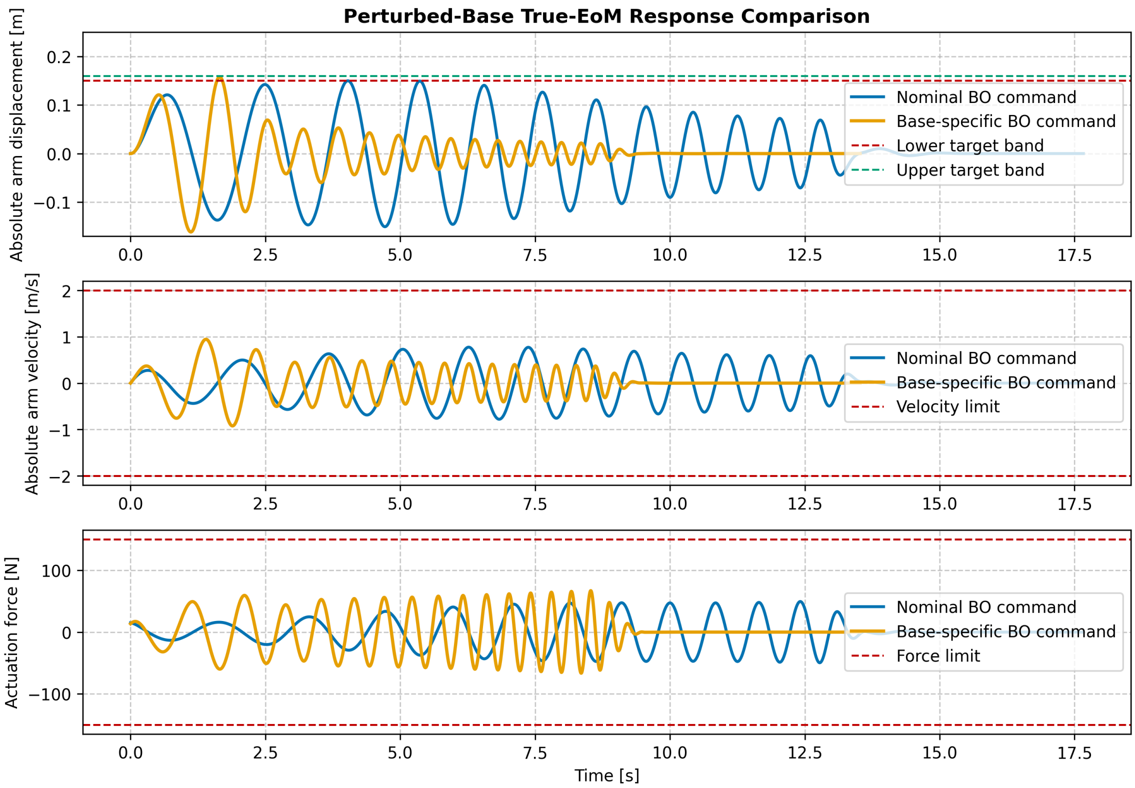

Figure 17: True-EoM response comparison on the selected perturbed base. The nominal v11 command produces a target shortfall, while the base-specific BO candidate recovers the target band and remains within the velocity and force limits.

As in the nominal v11 optimization, the NN surrogate was used only to guide the BO search. Final acceptance was based on true-EoM validation. The comparison in Table 15 shows that this distinction is essential. For the final base-specific candidate, the NN predicts a small target shortfall, whereas the true EoM gives a target-band response. Therefore, the surrogate correctly guides the search toward a useful region, but the final decision must still be taken using the reference EoM.

Table 15: NN-surrogate prediction versus true-EoM validation for the selected base-specific BO candidate.

Quantity	**NN prediction**	**True EoM**	**Relative error [%]**
$f_{res,1}$ [Hz]	0.8120	0.8146	0.32
$f_{res,2}$ [Hz]	1.3342	1.2735	4.76
$\max(y)$ [m]	0.148343	0.157711	5.94
v_{max} [m/s]	0.9711	0.9449	2.77
F_{max} [N]	86.56	67.01	29.18
Maximum tracking error [m]	0.1591	0.1705	6.69

This result is important for interpreting the base-robustness campaign. The loss of admissibility of the nominal v11 command on the selected perturbed base does not imply that the perturbed system is infeasible. It shows instead that the nominal optimum is base-dependent. Re-running the same BO pipeline for the modified base recovers an admissible solution. At the same time, the NN–EoM discrepancy observed for the base-specific candidate confirms again that the surrogate should not be used as the final authority on constraint satisfaction.

6 Robustness validation: arm parameters perturbation

A second robustness campaign was performed to assess the sensitivity of the fixed v11 command to local perturbations of the equivalent arm dynamics. The compliant base and the optimized chirp were kept fixed, while the equivalent arm stiffness, damping ratio and Cartesian mass were perturbed. Differently from the base-parameter campaign, this is not a broad stress test: it is a local full-factorial validation around the arm configuration associated with the final BO candidate.

All cases were evaluated using the true EoM, as in the base-parameter robustness validation. The reason is the same: this stage is not a fast design-space exploration step, but an accuracy-oriented validation of one already selected command. The NN surrogate is therefore not used to decide whether a perturbed configuration is admissible.

6.1 Local full-factorial definition

The nominal arm configuration used as the centre of the local perturbation grid is reported in Table 16. The equivalent arm stiffness and damping ratio correspond to the final v11 BO candidate, while the Cartesian arm mass is the nominal value used in the model.

Table 16: Nominal arm parameters used as the centre of the local full-factorial validation.

Parameter	Meaning	Value	Unit
m_a^*	Equivalent arm mass	10.0	kg
k_a^*	Equivalent arm stiffness	289.018	N/m
h_a^*	Equivalent arm damping ratio	0.5085	–

The perturbed quantities are expressed through the non-dimensional ratios

$$\frac{k_a}{k_a^*}, \qquad \frac{h_a}{h_a^*}, \qquad \frac{m_a}{m_a^*}. \tag{56}$$

The tested levels were

$$\frac{k_a}{k_a^*} \in \{0.95,\ 0.975,\ 1.00,\ 1.025,\ 1.05\}, \tag{57}$$

$$\frac{h_a}{h_a^*} \in \{0.90,\ 0.95,\ 1.00,\ 1.05,\ 1.10\}, \qquad \frac{m_a}{m_a^*} \in \{0.90,\ 0.95,\ 1.00,\ 1.05,\ 1.10\}. \tag{58}$$

The resulting campaign contains

$$5 \times 5 \times 5 = 125 \tag{59}$$

true-EoM simulations. The ranges are intentionally local and are not identical for the three parameters. They are intended to represent small identification or measurement uncertainties around the selected equivalent arm model, not a broad physical stress envelope. In particular, the stiffness is tested within a narrower ±5% interval, whereas damping ratio and equivalent mass are tested within ±10% intervals.

In the heatmaps, each panel corresponds to one fixed value of h_a/h_a^*, the horizontal axis reports m_a/m_a^* and the vertical axis reports k_a/k_a^*, consistently with the mass–stiffness layout used in the base-parameter robustness section.

6.2 Admissibility criteria

The same reaching-margin convention introduced in the base-parameter robustness section is used here. The target band is satisfied only if $0 \leq M_y \leq M_{allow}$: negative values indicate target shortfall, whereas values larger than M_{allow} indicate target overshoot. The velocity, force and dominant-phase margins must also remain non-negative. As before, numerical success only means that the time-domain integration completed, whereas admissibility refers to the fulfilment of all active design criteria.

6.3 Arm-parameter robustness results

All 125 integrations were numerically successful. The target band was satisfied in 75 cases, corresponding to 60.0% of the grid. All 50 non-admissible cases are target-shortfall cases. No target-overshoot, velocity, force or dominant-phase violation occurs in the tested local domain. Therefore, the arm-parameter campaign identifies a clean limiting mechanism: increasing the equivalent arm damping reduces the dynamic amplification until the lower target is no longer reached.

The reaching margin ranges from

$$M_y \in [-0.005\,92\,\mathrm{m},\ 0.008\,97\,\mathrm{m}]. \tag{60}$$

The minimum occurs at

$$\frac{k_a}{k_a^*} = 0.95, \qquad \frac{h_a}{h_a^*} = 1.10, \qquad \frac{m_a}{m_a^*} = 0.90, \tag{61}$$

whereas the maximum occurs at

$$\frac{k_a}{k_a^*} = 1.05, \qquad \frac{h_a}{h_a^*} = 0.90, \qquad \frac{m_a}{m_a^*} = 0.95. \tag{62}$$

The maximum positive reaching margin remains below the upper target-band allowance, so no excessive overshoot is observed in this local arm-parameter grid.

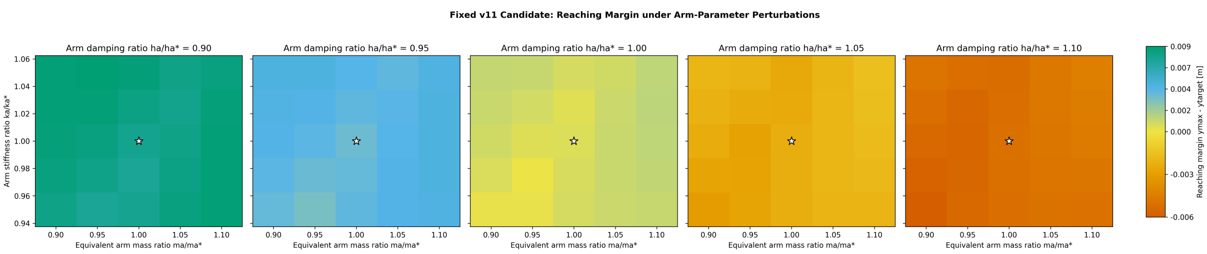

Figure 18: Reaching margin under local perturbations of the equivalent arm parameters. Positive values indicate that the lower target is reached.

The reaching-margin heatmap in Fig. 18 shows that the dominant trend is the strong reduction of reaching margin when the arm damping ratio increases. This is quantified by the Spearman coefficient $\rho(M_y, h_a/h_a^*) = -0.98$. The correlations with stiffness and equivalent mass are much weaker, $\rho(M_y, k_a/k_a^*) = +0.07$ and $\rho(M_y, m_a/m_a^*) = +0.10$. Hence, within the tested local range, the admissibility boundary is controlled almost entirely by damping.

The velocity limit is never active in the arm-parameter grid. The maximum observed velocity is

$$v_{max,max} = 0.861\,\mathrm{m/s}, \tag{63}$$

well below $v_{lim} = 2\,\mathrm{m/s}$.

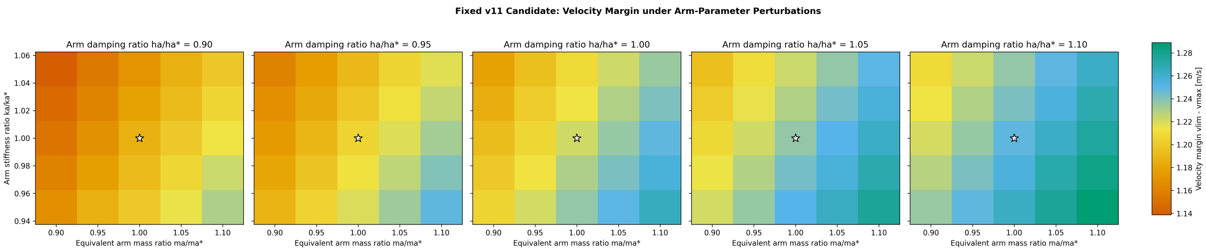

Figure 19: Velocity margin under arm-parameter perturbations. All tested configurations remain below the active 2 m/s velocity limit.

Since $M_v = v_{lim} - v_{max}$, a larger velocity margin means a less critical velocity response. The velocity margin increases with damping and equivalent mass, with $\rho = +0.71$ and $\rho = +0.62$, respectively. This is physically consistent with a more damped or heavier arm producing lower peak velocities. The stiffness correlation is weaker and negative, $\rho = -0.29$, indicating that increasing stiffness tends to slightly reduce the available velocity margin. In all cases, however, the velocity margin remains large.

The force limit is also never active. The maximum observed actuation force is

$$F_{max,max} = 51.91\,\mathrm{N}, \tag{64}$$

which is well below $F_{lim} = 150\,\mathrm{N}$.

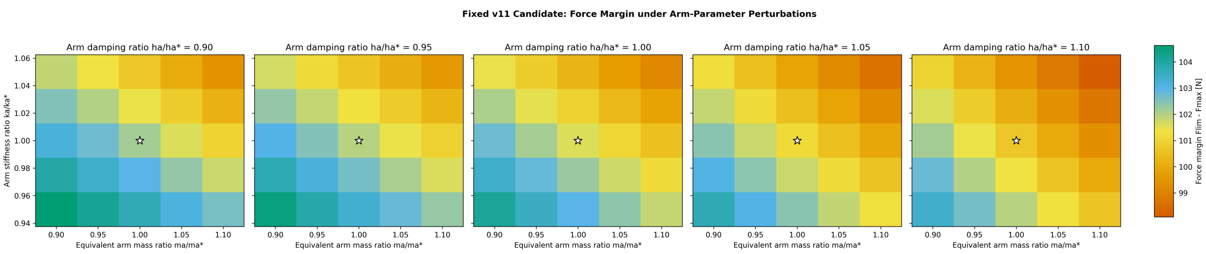

Figure 20: Force margin under arm-parameter perturbations. The force limit is respected throughout the full factorial grid.

The force margin decreases as stiffness, damping and mass increase, with Spearman coefficients $\rho(M_F, k_a/k_a^*) = -0.69$, $\rho(M_F, h_a/h_a^*) = -0.36$ and $\rho(M_F, m_a/m_a^*) = -0.61$. Thus, stiffer and heavier arm configurations tend to require higher actuation force. Nevertheless, the minimum force margin remains 98.09 N, so force is not a limiting factor for this campaign.

The dominant-phase criterion remains satisfied across the complete grid. The maximum absolute dominant phase difference is 28.57 deg, leaving a minimum phase margin of 61.43 deg.

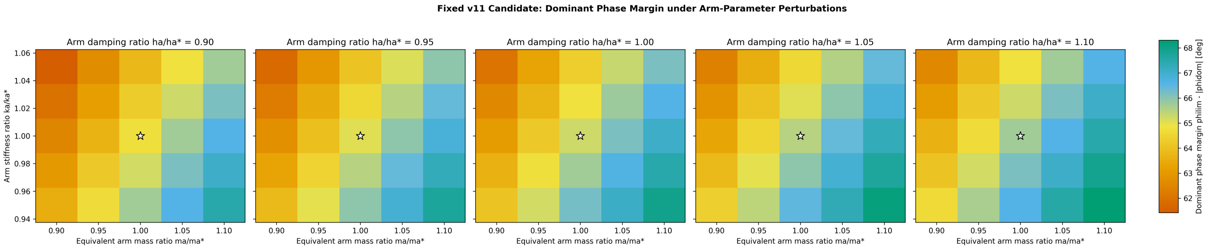

Figure 21: Dominant phase margin under arm-parameter perturbations. The phase criterion remains satisfied across the complete factorial domain.

The phase margin is mainly influenced by equivalent arm mass. The Spearman coefficient is $\rho(M_\phi, m_a/m_a^*) = +0.89$, meaning that increasing the equivalent mass improves the dominant-phase margin over the tested local range. The stiffness effect is moderately negative, $\rho(M_\phi, k_a/k_a^*) = -0.40$, while the damping effect is weakly positive, $\rho(M_\phi, h_a/h_a^*) = +0.20$. Since all phase margins remain far from zero, this trend is diagnostic rather than limiting.

The category map in Fig. 22 summarizes the admissibility outcome. All cases with $h_a/h_a^* \leq 1.00$ are admissible, whereas all cases with $h_a/h_a^* \geq 1.05$ fail by target shortfall. Stiffness and equivalent mass affect the margin values, but they do not change the admissibility outcome once the damping level is fixed.

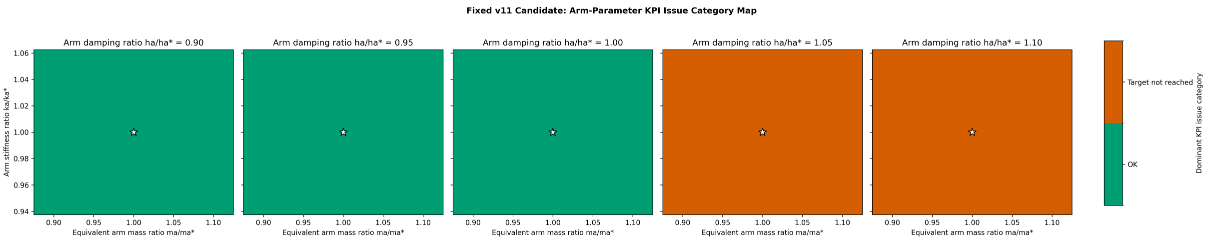

Figure 22: Dominant KPI issue category under arm-parameter perturbations. Target shortfall appears only for increased damping levels.

The violation summary in Fig. 23 confirms that target shortfall is the only active violation mode in the arm-parameter campaign.

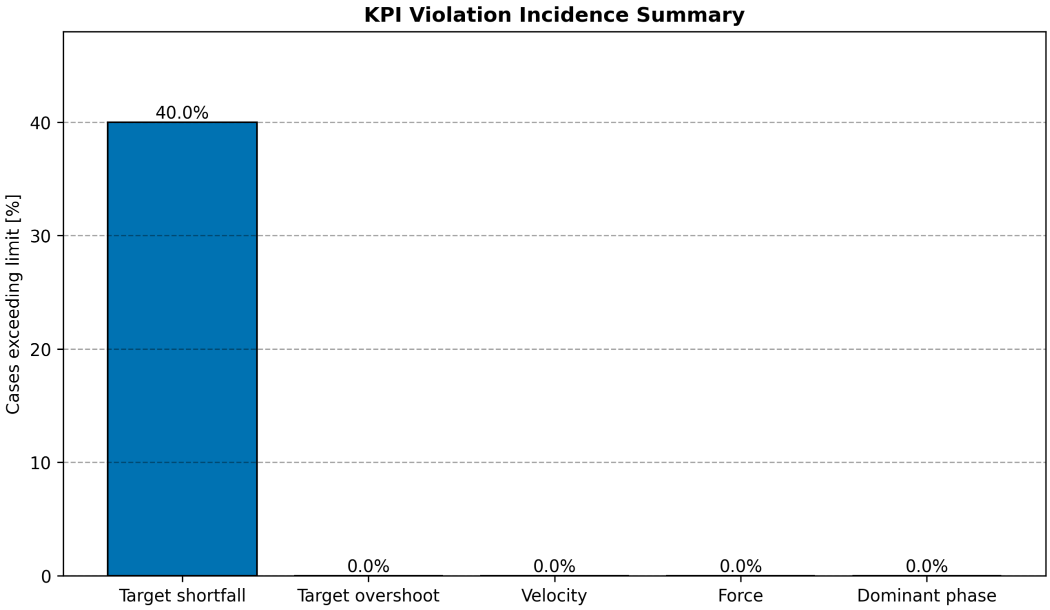

Figure 23: Percentage of arm-parameter simulations violating each monitored criterion. Only target shortfall occurs.

Table 17: Summary of the fixed v11 arm-parameter robustness campaign.

Criterion	Cases	Fraction [%]
Numerically successful simulations	125	100.0
Target band satisfied / admissible	75	60.0
Target shortfall	50	40.0
Target overshoot	0	0.0
Velocity limit exceeded	0	0.0
Force limit exceeded	0	0.0
Dominant phase limit exceeded	0	0.0

6.4 FRF and sensitivity interpretation

A one-at-a-time FRF analysis was performed by perturbing the arm stiffness, damping ratio and equivalent mass separately. The fixed v11 chirp window remains

$$[f_{min}, f_{max}] = [0.436\,\mathrm{Hz}, 1.347\,\mathrm{Hz}]. \tag{65}$$

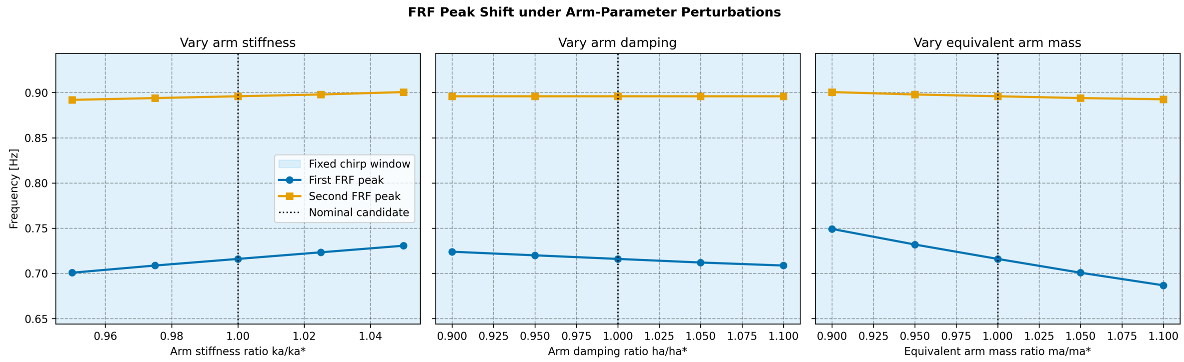

Figure 24: One-at-a-time displacement of the first and second FRF peaks under arm-parameter perturbations. Both peaks remain inside the fixed v11 chirp window over the tested local domain.

Figure 24 shows that the relevant modal peaks remain inside the fixed chirp window for all tested local arm-parameter perturbations. Therefore, the loss of admissibility is not caused by a resonance moving outside the excitation band. Instead, the heatmaps and the Spearman coefficient indicate a damping-induced attenuation mechanism: increasing h_a dissipates more of the chirp-induced motion and reduces the maximum absolute displacement reached by the arm.

As in the base-parameter validation, the Spearman rank-correlation coefficient is used as a compact indicator of monotonic trends over the full grid. It should be interpreted together with the heatmaps, because it does not replace the full factorial maps and does not capture all parameter interactions.

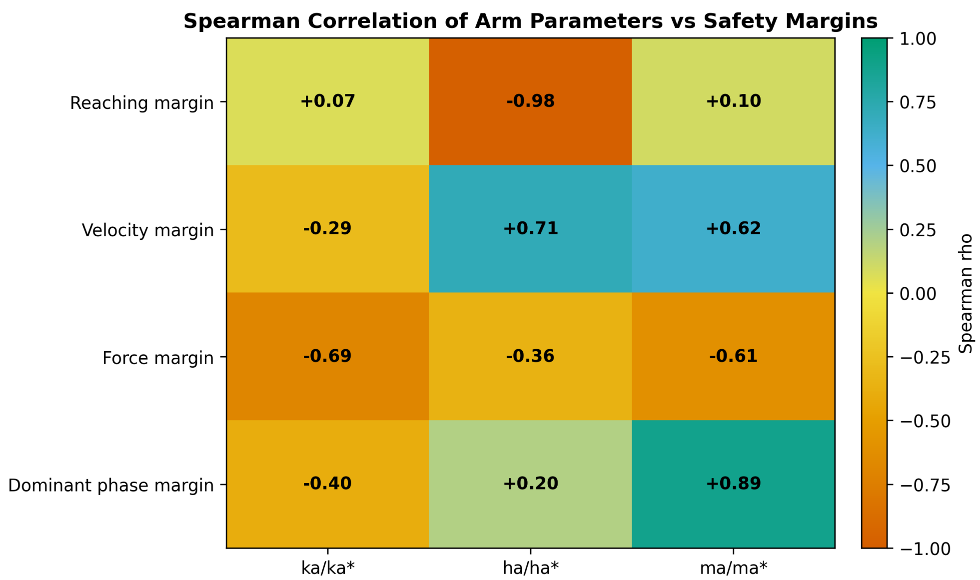

Figure 25: Spearman correlation between the perturbed arm parameters and the safety margins. Arm damping is the dominant parameter for the reaching margin.

The Spearman map in Fig. 25 confirms that damping dominates the reaching margin, while the other margins follow different secondary trends. Velocity margin increases with damping and mass, force margin decreases mainly with stiffness and mass, and phase margin improves strongly with equivalent mass. These trends are useful for interpretation, but they do not change the admissibility conclusion: the only active failure mechanism in the tested arm-parameter domain is target shortfall caused by increased damping.

6.5 Verified arm validity domain

The verified all-admissible arm-parameter box is

$$\boxed{\frac{k_a}{k_a^*} \in [0.95, 1.05], \quad \frac{h_a}{h_a^*} \in [0.90, 1.00], \quad \frac{m_a}{m_a^*} \in [0.90, 1.10]}. \tag{66}$$

All tested combinations inside this box satisfy the target band, velocity, force and dominant-phase criteria. As for the base-parameter validation, this box is a discrete verified domain obtained from the tested grid and should not be interpreted as a continuous guarantee between sampled points. The true boundary in damping lies between $h_a/h_a^* = 1.00$ and $h_a/h_a^* = 1.05$, because all cases at 1.05 already fail by target shortfall. A finer grid could refine this threshold, but it is not necessary for the present report because the controlling mechanism is already clear.

7 Robustness validation: Initial conditions perturbation

After validating the fixed v11 command against base and arm parameter perturbations, a final robustness campaign was performed by perturbing the initial state of the system. The objective of this analysis is different from the previous parameter-robustness tests: the mechanical model and the optimized chirp are kept fixed, while the initial displacement and velocity conditions are varied. Therefore, the campaign checks whether the optimized command remains admissible when the motion starts from a non-zero state.

7.1 Initial-condition perturbation setup

All cases were evaluated with the true EoM. As in the base- and arm-parameter robustness sections, this stage is an accuracy-oriented validation of the selected command, not a fast design-space exploration step. The NN surrogate is therefore not used to classify the initial-condition cases.

The initial state was defined using physically interpretable absolute coordinates:

$$y_0, \qquad x_{b0}, \qquad \dot{y}_0, \qquad \dot{x}_{b0}, \tag{67}$$

where y_0 is the absolute arm position, x_{b0} is the absolute base position, $\dot{y}_0$ is the absolute arm velocity and $\dot{x}_{b0}$ is the absolute base velocity. These quantities are then converted into the coordinates used by the EoM solver as

$$x_{a0} = y_0 - x_{b0}, \qquad \dot{x}_{a0} = \dot{y}_0 - \dot{x}_{b0}. \tag{68}$$

This choice avoids the ambiguity of perturbing only the relative arm coordinate x_a, because the target is defined on the absolute arm displacement $y = x_a + x_b$.

A single initial-condition severity index s_{IC} was introduced to keep the grid compact and readable. For each severity level, the position perturbation scale P_s and the velocity perturbation scale V_s are defined as

$$P_s = \alpha_s A, \qquad V_s = \beta_s \dot{x}_{ref}(0), \tag{69}$$

where A is the optimized v11 chirp amplitude and $\dot{x}_{ref}(0)$ is the initial velocity of the imposed chirp reference. The position scale is therefore expressed relative to the amplitude of the command applied to the arm, while the velocity scale is expressed relative to the initial commanded velocity. With the v11 candidate,

$$A = 0.099\,756\,\mathrm{m}, \tag{70}$$

while

$$\dot{x}_{ref}(0) = 2\pi f_{min} A = 2\pi\,(0.436\,132\,\mathrm{Hz})\,(0.099\,756\,\mathrm{m}) \simeq 0.273\,\mathrm{m/s}. \tag{71}$$

This velocity scale was selected because a velocity perturbation of order $\dot{x}_{ref}(0)$ corresponds to an initial motion of the same order as the initial commanded chirp velocity.

Table 18 reports the adopted severity levels in terms of the non-dimensional scaling factors α_s and β_s. The corresponding physical values are not used as the primary grid labels, because the purpose of the index is to provide a compact severity scale for the plots.

Table 18: Definition of the initial-condition severity levels. The position scale is expressed with respect to the optimized chirp amplitude A, while the velocity scale is expressed with respect to the initial chirp-reference velocity $\dot{x}_{ref}(0)$.

Severity level s_{IC}	**Position scale** P_s	**Velocity scale** V_s
1	$0.005A$	$0.25\dot{x}_{ref}(0)$
2	$0.010A$	$0.50\dot{x}_{ref}(0)$
3	$0.025A$	$1.00\dot{x}_{ref}(0)$
4	$0.100A$	$2.00\dot{x}_{ref}(0)$
5	$0.500A$	$3.00\dot{x}_{ref}(0)$

For each severity level, all sign and zero combinations were tested:

$$y_0, x_{b0} \in \{-P_s, 0, +P_s\}, \qquad \dot{y}_0, \dot{x}_{b0} \in \{-V_s, 0, +V_s\}. \tag{72}$$

Thus, each level contains $3^4 = 81$ true-EoM simulations. The complete campaign contains

$$5 \times 81 = 405 \tag{73}$$

true-EoM integrations. The all-zero initial condition appears once for each severity level because it belongs to every severity grid. These repeated zero-state entries are kept in the plots to preserve the same matrix layout for all levels; consequently, the campaign contains 405 simulations but 401 unique initial states.

The same admissibility criteria introduced in the previous robustness sections are used here. The target band is satisfied only if the reaching margin lies in the interval $0 \leq M_y \leq M_{allow}$; negative values indicate target shortfall, while values larger than M_{allow} indicate target overshoot. A case is classified as admissible only if this target-band condition and the velocity, force and dominant-phase criteria are all satisfied.

7.2 Initial-condition robustness results

All 405 integrations completed successfully. Overall, 91.4% of the cases are admissible, while 8.6% violate the target band by excessive overshoot. No target shortfall, velocity violation, force violation or dominant-phase violation is observed. Therefore, the initial-condition campaign identifies a different limitation from the arm-parameter validation: the critical cases do not lose reachability, but rather start with enough favorable initial energy to exceed the upper target-band bound.

The global outcome is shown in Fig. 26. The only observed non-admissible category is target overshoot.

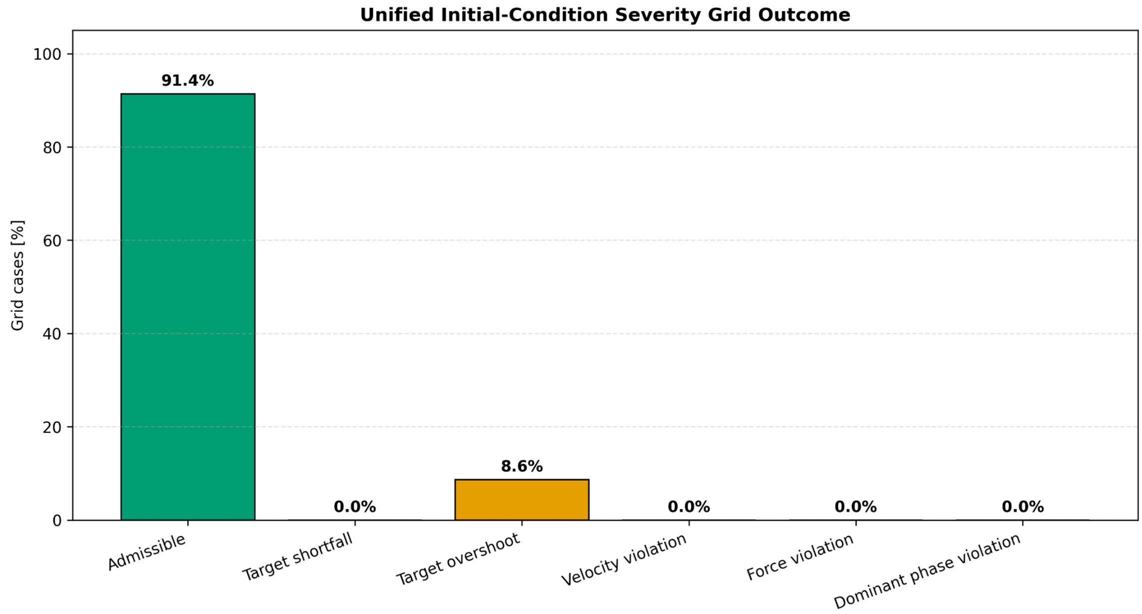

Figure 26: Global outcome of the unified initial-condition severity grid. The only observed non-admissible category is target overshoot.

Table 19: Summary of the fixed v11 initial-condition robustness campaign.

Criterion	**Cases**	**Fraction [%]**
Numerically successful simulations	405	100.0
Target band satisfied / admissible	370	91.4
Non-admissible cases	35	8.6
Target shortfall	0	0.0
Target overshoot	35	8.6
Velocity limit exceeded	0	0.0
Force limit exceeded	0	0.0
Dominant-phase limit exceeded	0	0.0

The complete outcome matrix is reported in Fig. 27. The figure is intentionally interpreted only at a high level: the first three severity levels are fully admissible, while failures appear only for the two highest levels. This confirms that the v11 candidate has a well-defined admissible region with respect to the adopted initial-condition severity index, rather than failing uniformly over the tested domain.

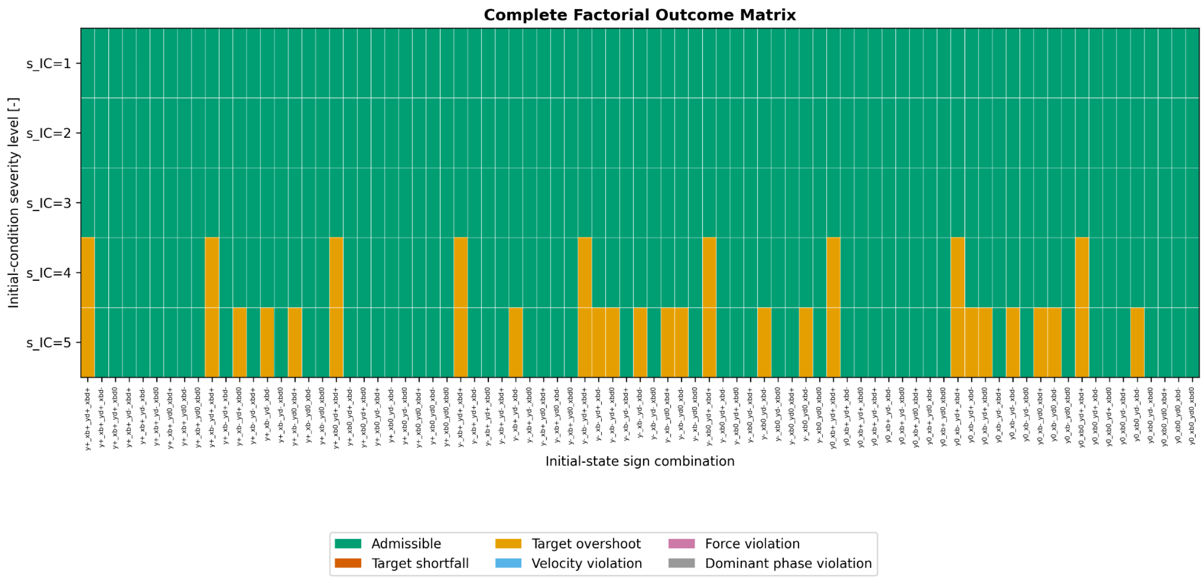

Figure 27: Complete factorial outcome matrix for the initial-condition grid. Failures appear only at the highest severity levels and are associated with target overshoot.

Figure 28 summarizes the same trend more clearly. Severity levels 1, 2 and 3 are fully admissible. At $s_{IC} = 4$, 11.1% of the cases fail, while at $s_{IC} = 5$ the failure fraction increases to 32.1%. This provides a compact validity statement:

$$\boxed{s_{IC} \leq 3 \quad \Rightarrow \quad \text{all tested initial-condition combinations are admissible.}} \tag{74}$$

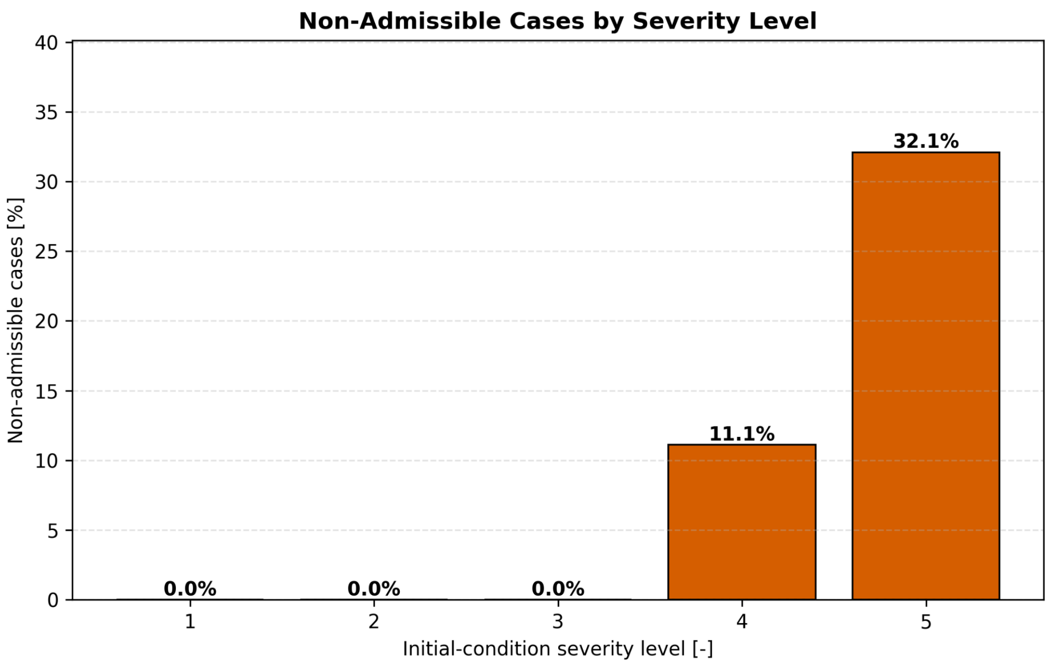

Figure 28: Non-admissible cases as a function of the initial-condition severity level. The first failures appear at $s_{IC} = 4$ and increase at $s_{IC} = 5$.

The problematic cases have a clear physical interpretation. At severity level 4, all non-admissible cases include

$$\dot{y}_0 > 0, \qquad \dot{x}_{b0} > 0, \tag{75}$$

meaning that both the absolute arm and the base initially move in the positive direction, i.e., toward the target side of the motion. This initial kinetic contribution adds to the subsequent chirp-induced amplification and can push the peak displacement above the upper target-band limit. At severity level 5, the overshoot region becomes broader.

The reaching-margin plot in Fig. 29 must be interpreted carefully. The minimum reaching margin remains positive for every severity level, decreasing only slightly from approximately 0.601 mm to 0.580 mm. This means that the lower target bound is always reached. However, a positive reaching margin is not sufficient for admissibility, because the target band also imposes an upper bound. The non-admissible cases are therefore not shortfall cases, but excessive-overshoot cases.

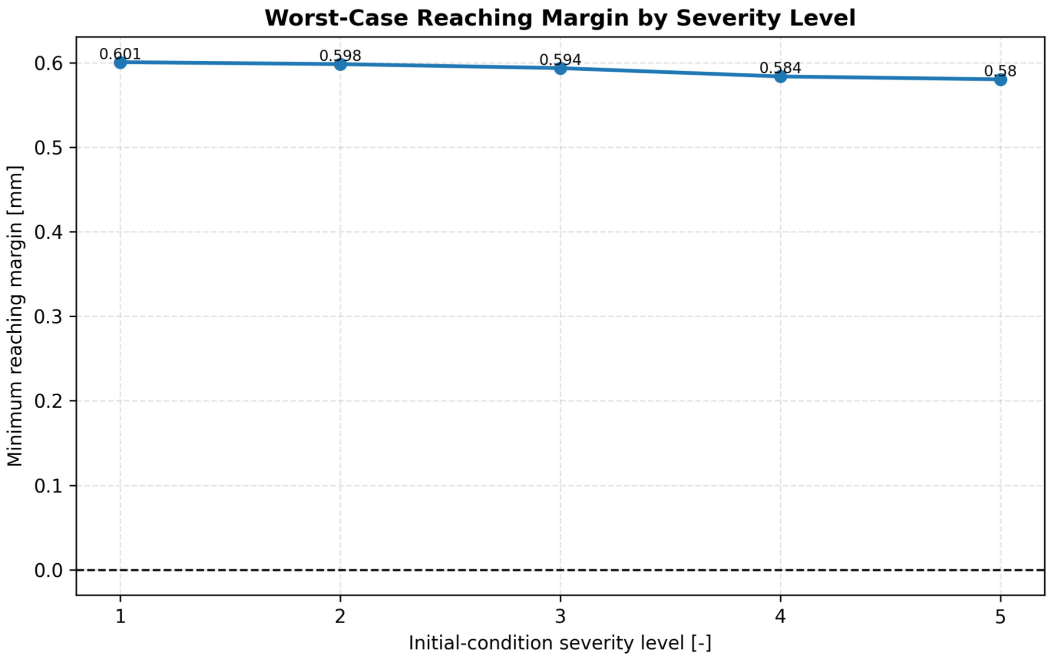

Figure 29: Worst-case reaching margin as a function of severity level. The lower target is reached in all tested cases; failures are instead caused by exceeding the upper target-band bound.

The force and velocity trends are reported in Figs. 30 and 31. Both quantities increase mainly at the highest severity levels, but neither becomes an active constraint. The maximum force observed over the grid is 133.45 N, still below the 150 N limit. The maximum absolute arm velocity is 1.001 m/s, about half of the 2 m/s limit. Therefore, the velocity and force responses remain below the adopted monitored limits in the tested initial-condition domain; the limiting mechanism is target-band overshoot.

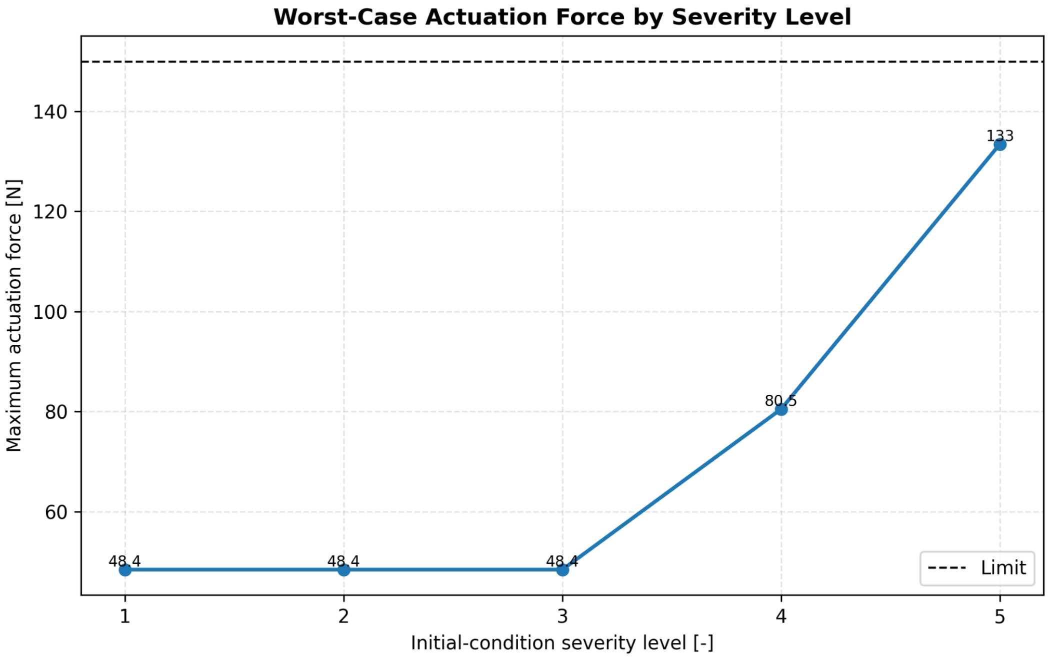

Figure 30: Worst-case actuation force as a function of severity level. The force increases for large initial-condition perturbations, but remains below the adopted limit.

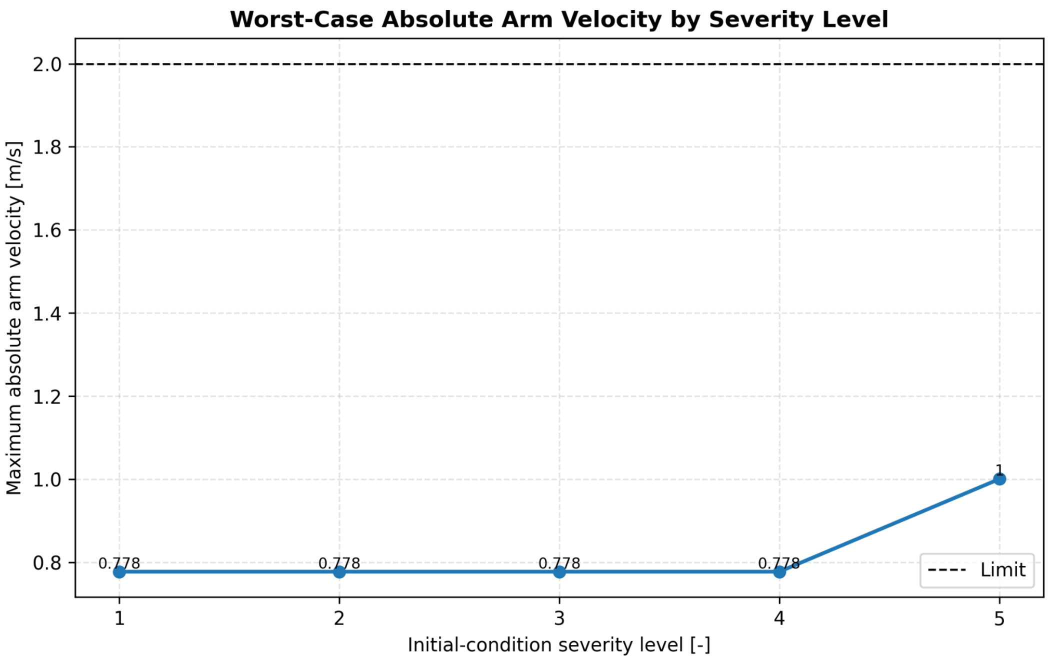

Figure 31: Worst-case absolute arm velocity as a function of severity level. The velocity constraint is respected in all tested initial-condition cases.

7.3 Worst-case analysis

The worst target-overshoot case is reported in Table 20. In this case, the arm absolute position starts already on the target side, while the base starts in the opposite direction. This creates a large initial relative arm displacement $x_{a0} = y_0 - x_{b0}$. In addition, both $\dot{y}_0$ and $\dot{x}_{b0}$ are positive and large. Although the initial relative velocity is zero in this specific case, the absolute motion starts in the same direction as the desired amplification. The resulting trajectory exceeds the upper target-band bound, reaching 0.234 66 m.

Table 20: Worst initial-condition case found in the unified severity grid. This case is selected as the largest target overshoot among all tested simulations.

Quantity	**Value**
Severity level s_{IC}	5
Sign combination	$y^+\ x_b^-\ \dot{y}^+\ \dot{x}_b^+$
y_0	+49.878 mm
x_{b0}	−49.878 mm
$x_{a0} = y_0 - x_{b0}$	+99.756 mm
$\dot{y}_0$	+0.820 m/s
$\dot{x}_{b0}$	+0.820 m/s
$\dot{x}_{a0} = \dot{y}_0 - \dot{x}_{b0}$	0.000 m/s
Maximum absolute arm displacement $\max(y)$	0.234 66 m
Maximum absolute arm velocity	0.820 m/s
Maximum actuation force	48.43 N
Failure category	Target overshoot

The time-domain comparison in Fig. 32 shows the nominal BO v11 response and the selected constraint-violating case. The baseline response reaches the target band with a small positive margin, while the worst case crosses the upper target bound during the initial transient. The velocity and force limits are still respected, confirming that the non-admissibility is caused by target overshoot only.

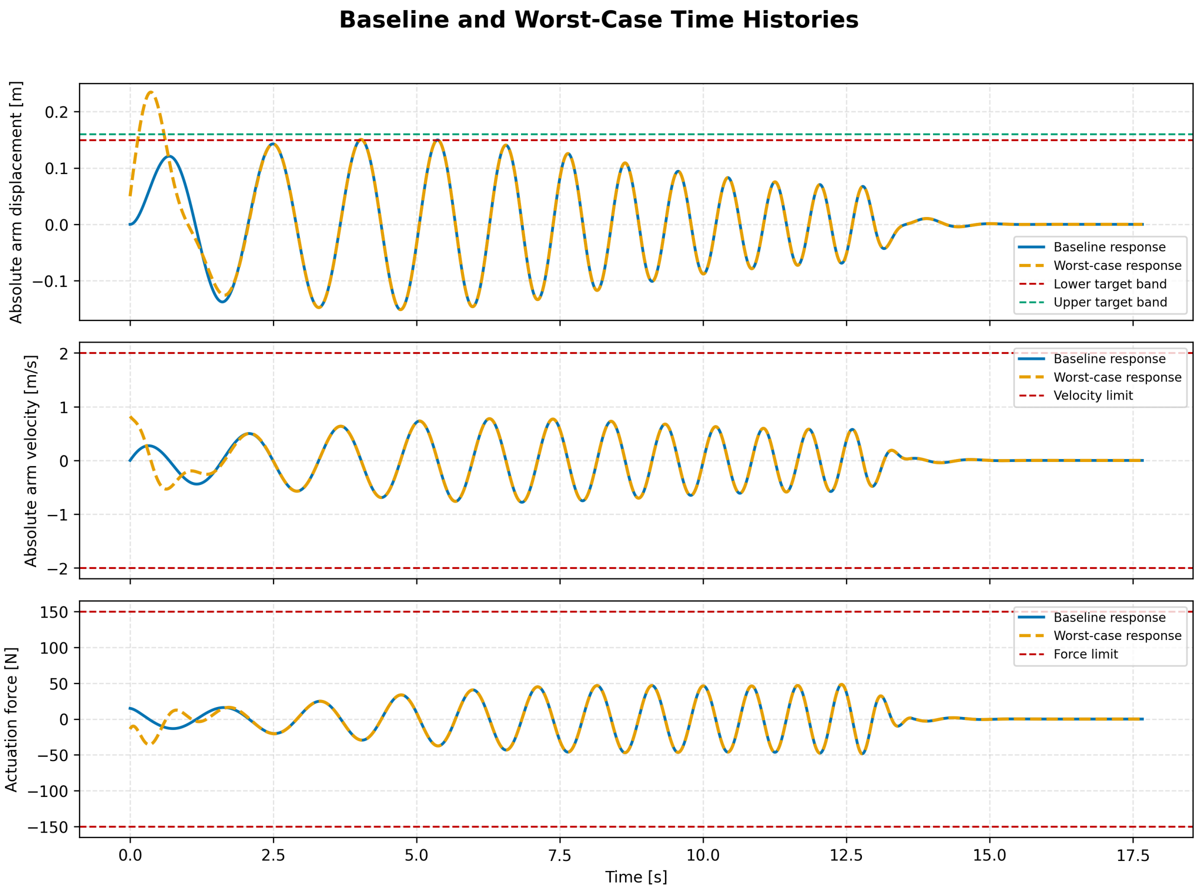

Figure 32: Baseline BO v11 response and worst constraint-violating initial-condition case. The failure is caused by target overshoot, while velocity and force remain below their limits.

The initial-condition analysis therefore refines the interpretation of the v11 candidate. The solution is robust to all tested initial-condition combinations up to severity level 3, corresponding to moderate perturbations of both position and velocity. At more aggressive severity levels, the fixed chirp can become too effective when the initial state already provides positive motion or displacement toward the target. This confirms that the v11 candidate is mild with respect to velocity and force, but its target-band admissibility still depends on avoiding large favourable initial-condition perturbations.

8 Conclusions

This project developed and validated a surrogate-assisted optimization workflow for a compliant base–arm system commanded through a chirp reference. The objective was not only to maximize displacement amplification, but to obtain a controlled target-band response: the absolute displacement $y = x_a + x_b$ had to reach the prescribed target while avoiding excessive overshoot and while keeping velocity, force and dominant-phase indicators within the adopted limits. The final formulation therefore combines a neural-network surrogate, Bayesian Optimization and true-equation-of-motion validation, rather than relying on any single component of the pipeline in isolation.

The neural-network surrogate proved suitable for accelerating the BO loop. A standard feed-forward NN was selected as the final surrogate because it provided accurate predictions on the BO-relevant KPIs and remained simpler and more robust than the tested physics-informed alternative. The PINN assessment was useful as an intermediate methodological check, but the additional physics-loss term did not produce a systematic per-output improvement on the reduced comparison dataset. For this reason, the traditional NN was retained, while all final candidates and robustness analyses were validated with the true EoM.

The final BO run produced a nominal chirp command that reaches the target band with high precision. The true-EoM validation gave $\max(y) = 0.150\,604\,\mathrm{m}$ for a target of $0.150\,000\,\mathrm{m}$, with velocity and force remaining below their monitored limits. The FRF and phase diagnostics confirmed that the selected chirp exploits a meaningful amplification region rather than a purely numerical artefact. At the same time, the small positive reaching margin made it necessary to test robustness explicitly instead of assuming that nominal feasibility implied robust feasibility.

The robustness campaigns clarified the limits of the nominal command. Base-parameter perturbations showed that the fixed v11 command is sensitive to changes in the compliant base dynamics: only 337 out of 845 tested base configurations were admissible, with target shortfall as the dominant failure mode and target overshoot as a secondary one. The subsequent base-specific re-optimization showed, however, that this limitation is mainly a limitation of the nominal command, not necessarily of the perturbed physical system: a new chirp optimized for the selected perturbed base recovered admissible target-band behaviour.

The arm-parameter robustness study showed a more local and more specific sensitivity. In that case, 75 out of 125 configurations were admissible, and all non-admissible cases were caused by target shortfall. The dominant mechanism was damping-induced attenuation: increasing the arm damping reduced the useful amplification without moving the relevant FRF peaks outside the chirp window. Finally, the initial-condition robustness campaign showed that the command is robust to moderate initial-state perturbations. All cases up to severity level $s_{IC} = 3$ were admissible, while the non-admissible cases at higher severity were caused only by target overshoot, not by force or velocity violations.

Overall, the final pipeline achieved the intended nominal target-band response and provided a clear map of where this response remains valid. The main limitation is that robustness is verified over selected grids and admissibility criteria, not proven globally.